19.07.2026 Leeture - 1

# Discrete Signals & Systems

**Signals:** A signal is a measurable physical quantity (such as voltage, pressure) that varies over an independent variable (like time or space) to carry meaningful information.

**Physically:** It is an observable parameter carrying meassage or data (e.g. sound waves, heartbeat traces).

**Mathematically:** It is expressed as a function of one ~~of~~ or more independent variables written as:

$x(t)$ ⟶ continuous time.

$x[n]$ ⟶ discrete "

**System:** ~~A signal is a measurable physical quantity~~

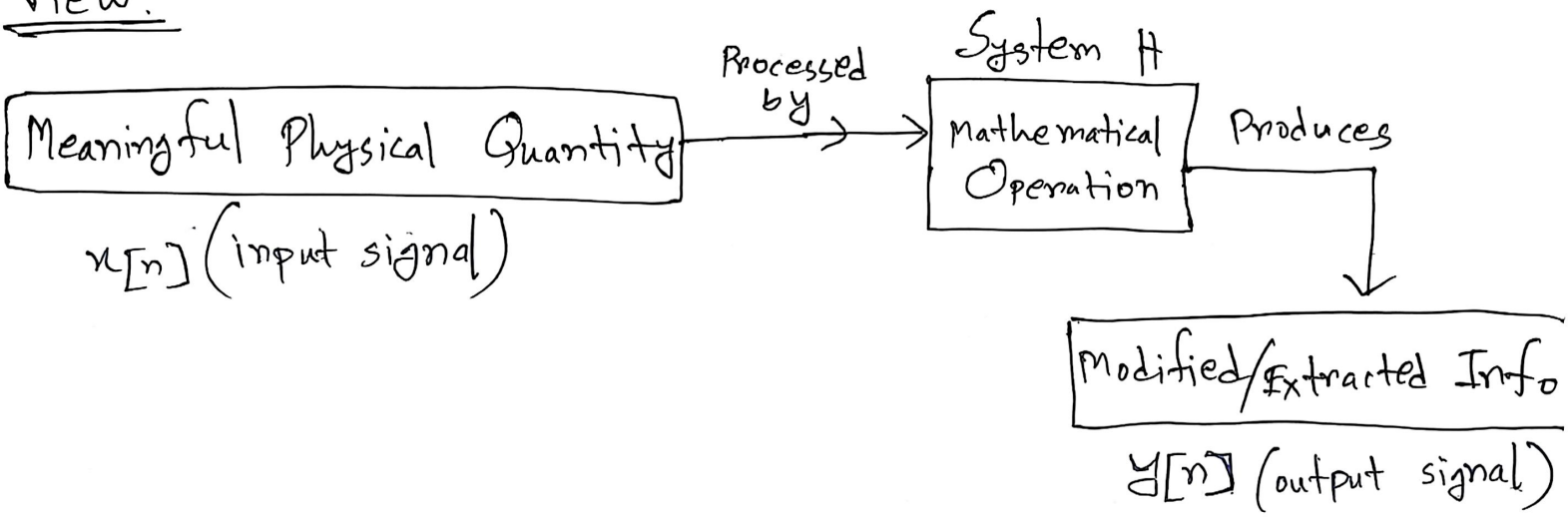
A system is set of interconnected components or an algorithm that process an input signal to extract, modify or transform its information to yield a specific output signal.

**Physically:** It is the entity or device doing the work (e.g. an amplifier, a noise-canceling circuit)

**Mathematically:** It represents a transform operator H that maps an input sequence $x[n]$ into an output sequence $y[n]$.

$$y[n] = H\{x[n]\}$$

**View:**




1.3

## Continuous and Discrete Time Signal

3 Types: ① Continuous Time Signal

② Discrete Time Signals.

③ Digital Signal

Audio Signal (A) → time → A(t)

Image (I) → space → I(s)

Video (v) → time, space → V(s, t)

① **Continuous-Time (CT) Signal:** A continuous-time signal is a physical or mathematical signal that is defined

continuously across a continuous range of time.

⇒ Mathematical Representation: It is denoted as $x(t)$, where the independent ~~signal~~ variable $t$ is a continuous variable. Here $(t \in \mathbb{R})$.

⇒ Key Characteristic: At Any exact instant of time t, no matter how ~~infinite~~ immeasurably small, $x(t)$ has defined value. for representation inside parenthesis ( ) as:

$x(t)$, $\delta(t)$, $u(t)$.

Examples: Human voice in air, body temperature throughout a day.

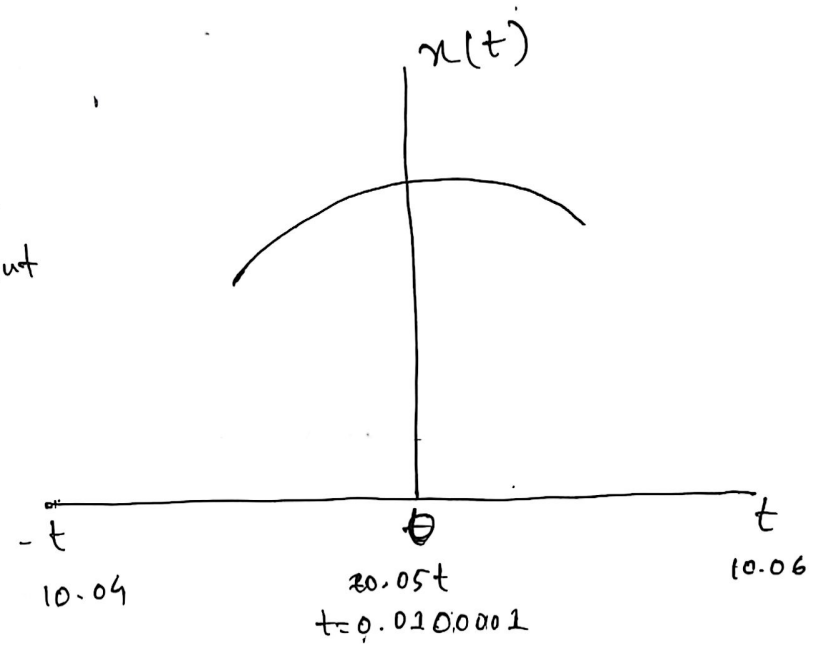



Fig. continuous-time signal

② Discrete-Time (DT) Signal: A Discrete-Time Signal is a signal that is defined only at discrete points in time. It is essentially a sequence of number indexed by an integer.

⇒ Mathematical Representation: Independent variable $n$, putting inside square paranthesis $[n]$. Denoted as $x[n]$, $g[n]$, $u[n]$ etc. $n$ is integer index ($n \in Z$) representing discrete time steps.

⇒ Key characteristics: $x[n]$ is undefined for non-integer values of $n$. For example $x[1.5]$ does not exist.

Examples: End-of-~~the~~day stock prices, monthly sales figures.

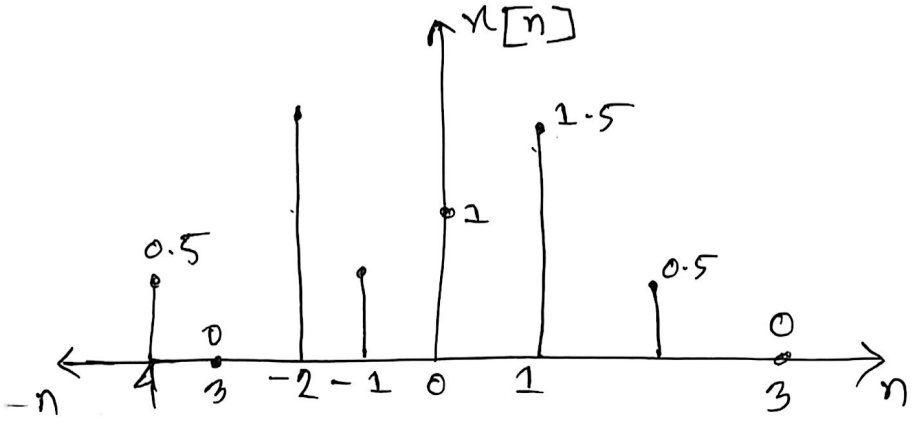
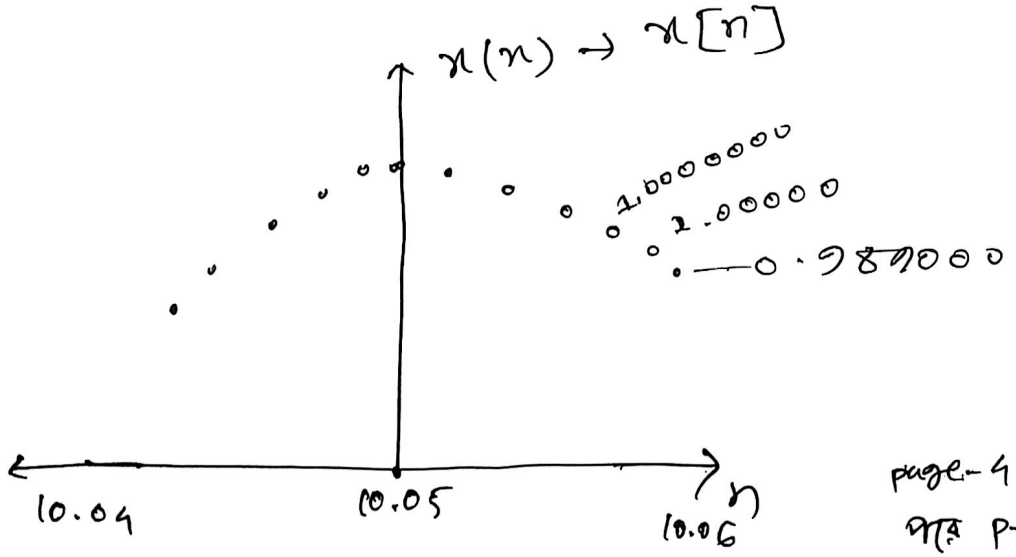



Fig: Discrete-Time Signal



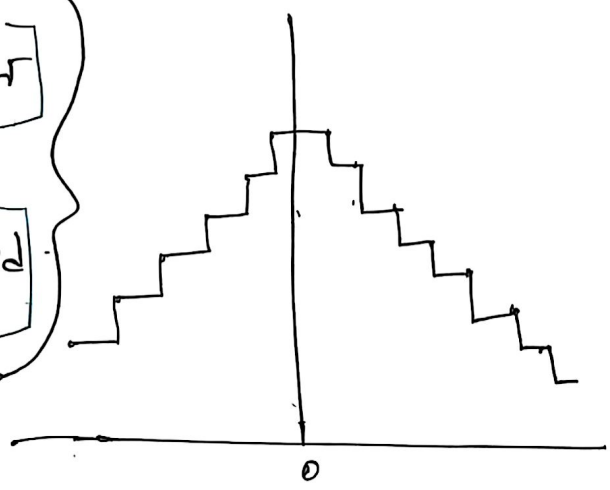

page-4 শেষ
পরে p-6 then p-5 //

(3) Digital Signal: A digital signal is a special type of discrete-time signal where both time and amplitude are discrete (quantized)

More precisely, a Digital Signal is a discrete-time signal whose values have been quantized so that a digital computer or processor can store and process it as binary numbers.

এমনের যতুন value না পাবে আগেরটা যেখানে।

Digital Signal এর value predefined কিন্তু কয়েকের infinity হবে না




Characteristics:

⊖ Sampling: The signal is evaluated only at specific integer intervals ($n = 0, 1, 2, \cdots$).

⊖ Quantization: The value/magnitude of the signal at each sample is rounded to the nearest level from a finite set of allowed values.

⊖ Representation: ~~Has~~ Usually represented in binary code (bits) inside computers, DSP chips and D.C. systems.

Digital ↓ comm.

# Discrete Signal Representation

1. Formula ⟶ $x[n] = \begin{cases} \left(\frac{1}{a}\right)^n, & n \geq 0 \\ 0, & n < 0 \end{cases}$

2. Sequence ⟶ $x[n] = \left\{ 0, 0, 0, \underset{\uparrow}{1}, \frac{1}{a}, \frac{1}{a^2}, \frac{1}{a^3}, \ldots \right\}$

$n=0$

arrow থাকলে সেই বরাবর $n=0$ শুরু হবে

3. Graphical Representation ⟶ $= \left\{ 1, \frac{1}{a}, \frac{1}{a^2}, \frac{1}{a^3} \right\}$

Note: যখন ⟶ sequence-এ কোনো অসম্ভব value বা ফাঁকা থাকবে তখন সেইটা 0 ধরে নিতে হবে।

⟶ এই প্রক্রিয়াকে zero padding বলে।

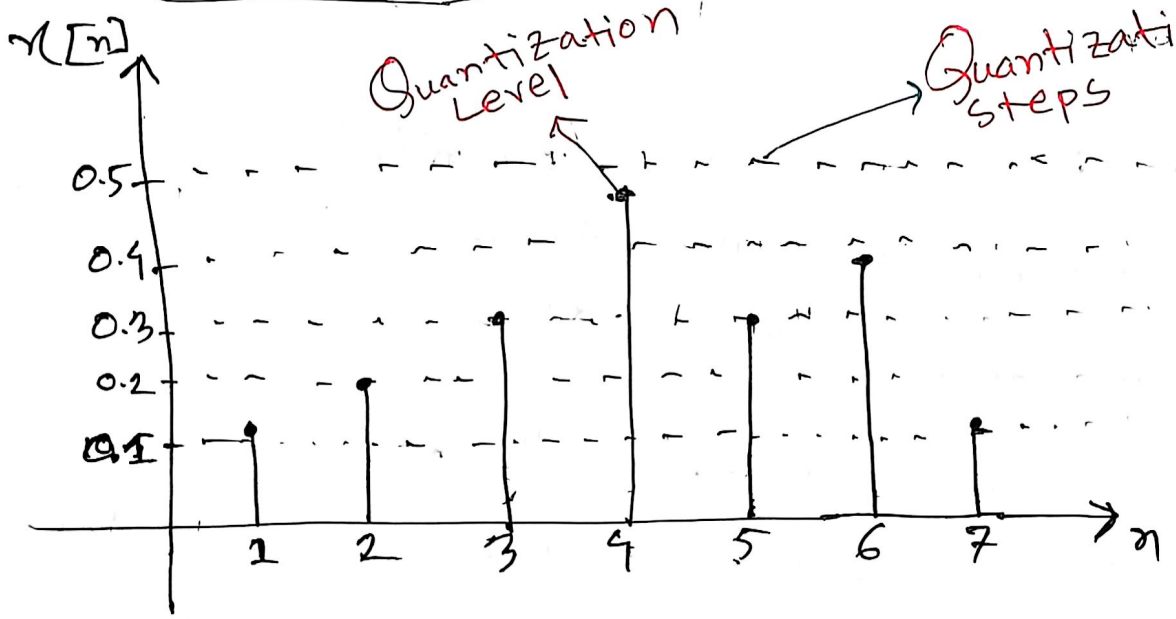



Fig: Digital Signal

## Book Examples

**Example:** Graphically represent $x[n] = \{3, 2, \underset{\uparrow}{5}, 4, 6, 8, 2\}$

**Sol$^n$:** Here, $x[n] = \{3, 2, \underset{\uparrow}{5}, 4, 6, 8, 2\}$

$$= \{\underset{n=-2}{3}, \underset{n=-1}{2}, \underset{n=0}{5}, \underset{n=1}{4}, \underset{n=2}{6}, \underset{n=3}{8}, \underset{n=4}{2}\}$$

$\therefore n = 0, 1, 2, 3, 4$

$x[0] = 5$

$x[1] = 4$

$x[2] = 6$

$x[3] = 8$

$x[4] = 2$

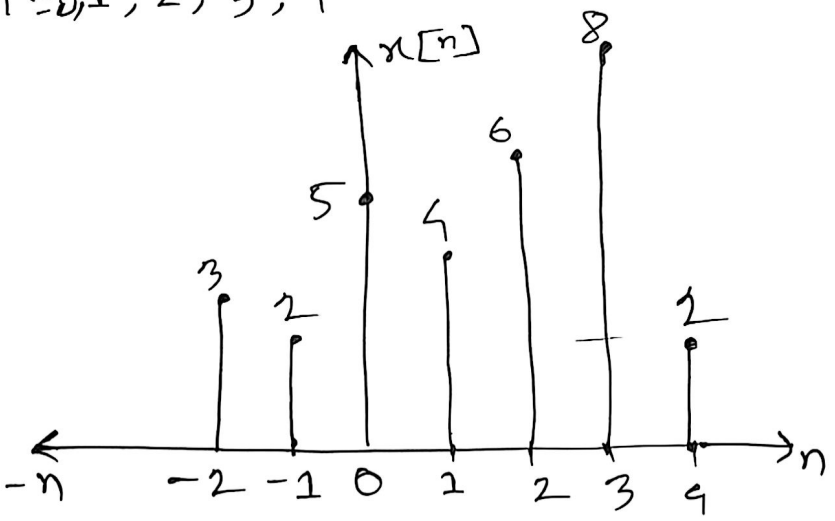



Fig: Graphical representation of $x[n]$

**Example: 1.1** Graphically represent $x[n] = \{1, 0, -1, 1\}$

Here, $x[n] = \{\underset{n=0}{1}, \underset{n=1}{0}, \underset{n=2}{-1}, \underset{n=3}{1}\}$

$\therefore n = 0, 1, 2, 3$

$x[0] = 1$

$x[1] = 0$

$x[2] = -1$

$x[3] = 1$

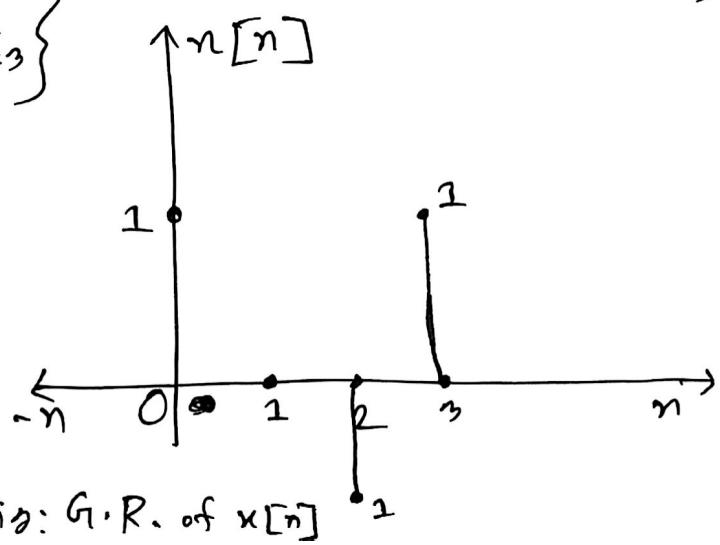



Fig: G.R. of $x[n]$

**Example- 1.2** Graphically represent the following sequence.

$$x[n] = \{-2, 1, 0, \underset{\uparrow}{1}, 2, 0, 1\}$$

22.07.2026 Lecture-2

# Digital Signal Processing

## 1.4 Basic Discrete Signal

### 1. Unit Impulse Signal

## CT-Signal vs DT-Signal

| Feature | Continuous-Time Signal | Discrete-Time Signal |
|---|---|---|
| Notation | $x(t)$ → uses round parenthesis | $x[n]$ → uses square bracket |
| Independent variable | $t$ → Continuous; real number | $n$ → Discrete, integer sequence |
| Domain | Defined at all time instants | Defined only at integer indices |
| Representation | Smooth continuous curves/waveforms | Sequence of discrete pulses/stems |
| Typical Context | Analog Circuits, Physical World | Computers, DSP microprocessors, digital algorithms |

## Key Distinction (Diff) (DT-Signal vs D.S.)

DT-Signal $x[n]$: Discrete in time, but amplitude can take any continuous real value (unquantized).

**D.S**: Discrete in time AND discrete in amplitude, that means quantized into a fixed set of levels, such as binary 0s and 1s.

## Summary

| Signal Type | Time Domain | Amplitude Domain | Examples |
|---|---|---|---|
| Analog (CT) | Continuous (t) | Continuous | Raw voice from a mic. |
| Discrete-Time (DT) | Discrete (n) | Continuous | Unquantized sampled voltage |
| Digital | Discrete (n) | Discrete (Quantized) | MP3 audio file, JPEG image bits. |

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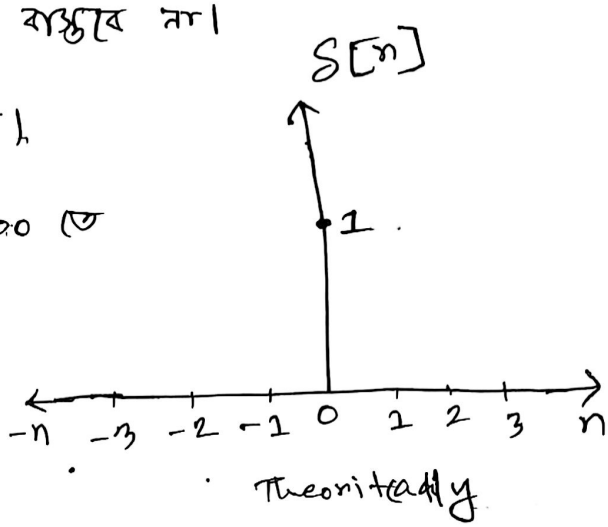
## 1.4 Basic Discrete Signal

Lecture-2
22.07.2026

1.4.1
**The Unit Impulse Sequence $\delta[n]$**: Also known as the Kronecker delta function, the unit impulse sequence is the most fundamental building block in discrete-time signal processing.

Impulse → Single pulse ↵
1 at n=0

~~একটি~~ 0 time -এ 1 থাকবে কিন্তু এটা বাস্তবে না।
(zero) duration
কারণ নির্দিষ্ট ~~সময়~~ পর্যন্ত 0 থাকে। যেমন: ঘড়ি।
59 second এর পরেই 01 মিনিট দেখায় না 00 তে
স্থির থাকে তারপর আবার 01.

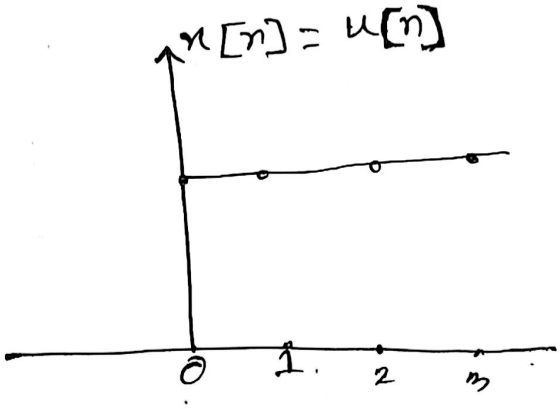


Theoritically

Mathematically: $$\delta[n] = \begin{cases} 1, & n = 0 \\ 0, & n \neq 0 \end{cases}$$

In short, the Unit Impulse Sequence denoted as $\delta[n]$, is a discrete signal that has a value of 1 at $n = 0$ and 0 everywhere else.

1.4.2 The Unit Step Sequence (u[n]): A discrete signal that turns ON (value = 1) at $n = 0$ and stays 1 for all positive values of n.




Usage: Models a digital switch turning on and staying on parmanently

Mathematically.

$$u[n] = \begin{cases} 1, & n \geq 0 \\ 0, & n < 0 \end{cases}$$

## 1.4.3 The Unit Ramp Signal ($r[n]$):

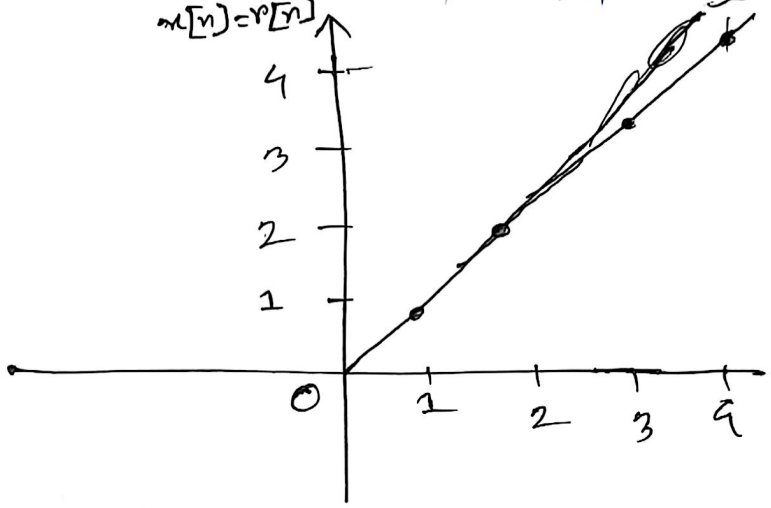
The Unit Ramp Sequence is signal whose amplitude grows linearly with discrete time $n$, starting at $n=0$.

value grows linearly with sample index $n$

Mathematically:

$$r[n] = \begin{cases} n, & n \geq 0 \\ 0, & n < 0 \end{cases}$$

or $r[n] = n \cdot u[n]$




Key idea: Values step up as; 0, 1, 2, 3, 4, .... for $n \geq 0$

## 1.4.6 Exponential Signal ($a^n$):

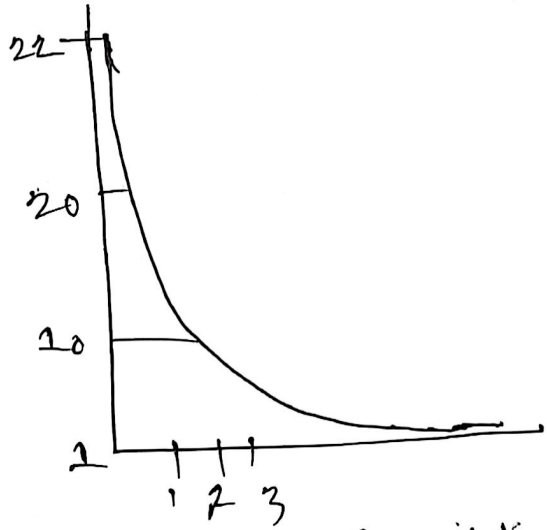
$Z = Ae^{\alpha t}$ represents a Continuous-Time Real Exponential Signal.

$Z = Ae^{\alpha t} \longrightarrow \alpha \pm$ (Gain)

$t \longrightarrow$ time

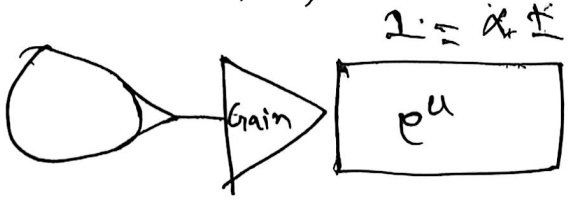
$A \longrightarrow$ Amplitude

change $Z(\alpha)$




Convert to D-T form:

$$x[n] = Z(nT_s)$$
$$= Ae^{\alpha(nT_s)}$$



$= A(e^{\alpha T_s})^n$

$= A a^n$ [Here: $a = e^{\alpha T_s}$]

∴ A sequence defined by $x[n] = a^n$, where its shape depends entirely on the base value a.

Note: → practical signal কখনই infinity হবে না বা 0 zero হবে না

$a > 1$ / $0 < a < 1$ / $a (|a < 0)$ :

## (1.5) Signal Operations on DT signal

Lecture-3
26.07.2026

1. Addition: Signal Addition is performed sample-by-sample (point-to-point) at identical time indices n. Consider the signals $x_1[n]$ and $x_2[n]$ and the addition of these signals at every n is represented as $x[n] = x_1[n] + x_2[n]$

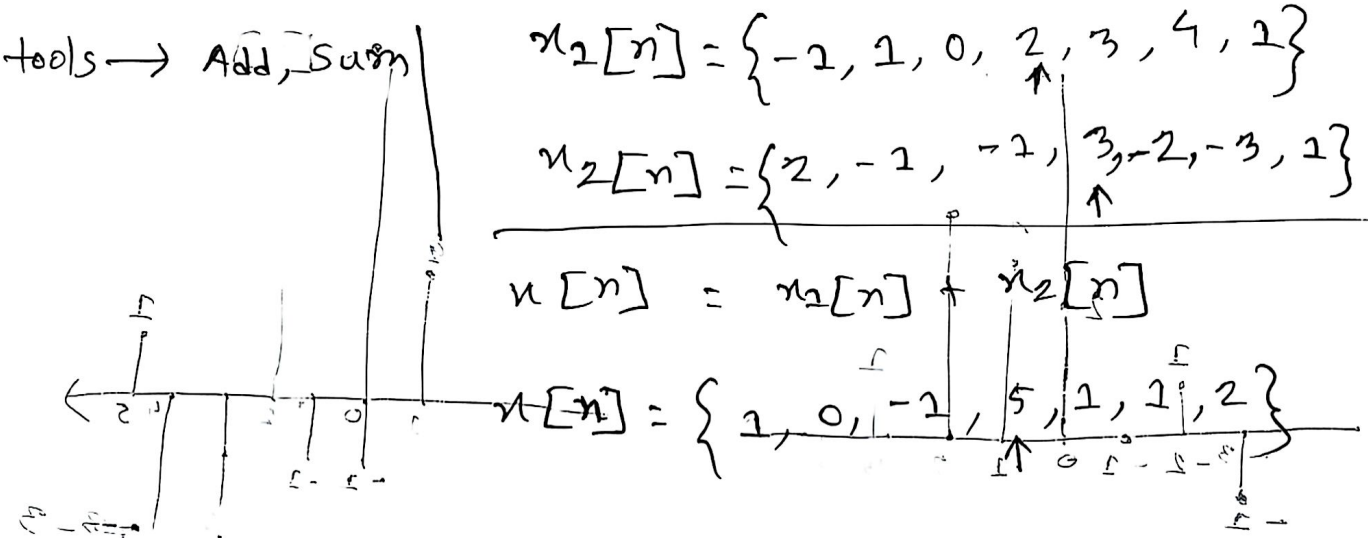
Matlab tools → Add, Sum

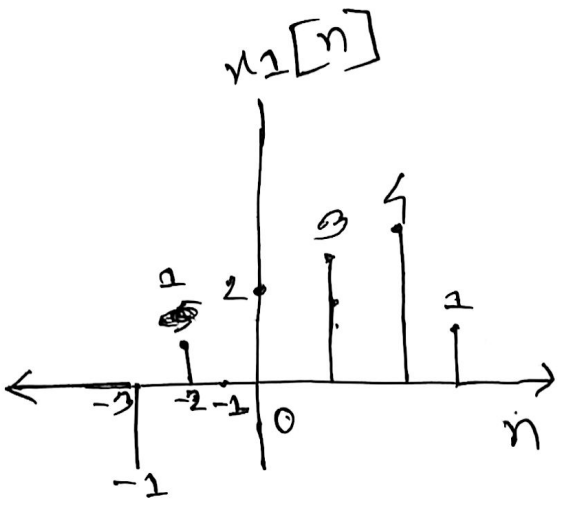
$x_1[n] = \{-1, 1, 0, \underset{\uparrow}{2}, 3, 4, 1\}$

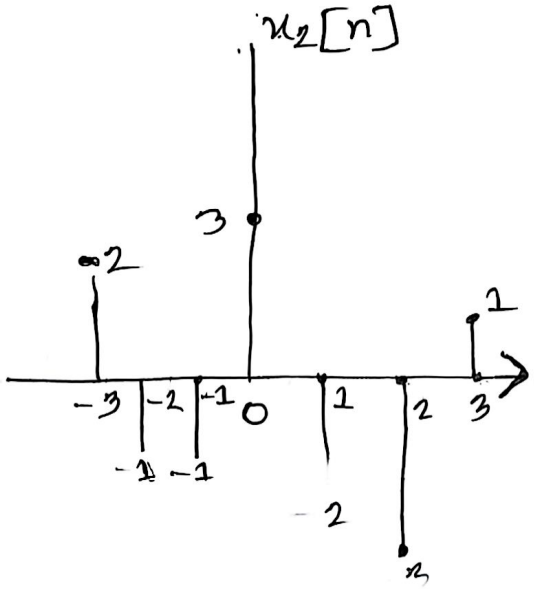
$x_2[n] = \{2, -1, -1, \underset{\uparrow}{3}, -2, -3, 1\}$

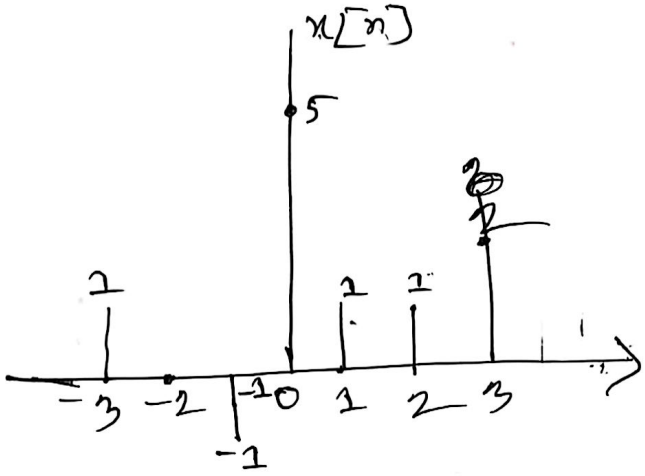
$x[n] = x_1[n] + x_2[n]$

$x[n] = \{1, 0, -1, \underset{\uparrow}{5}, 1, 1, 2\}$





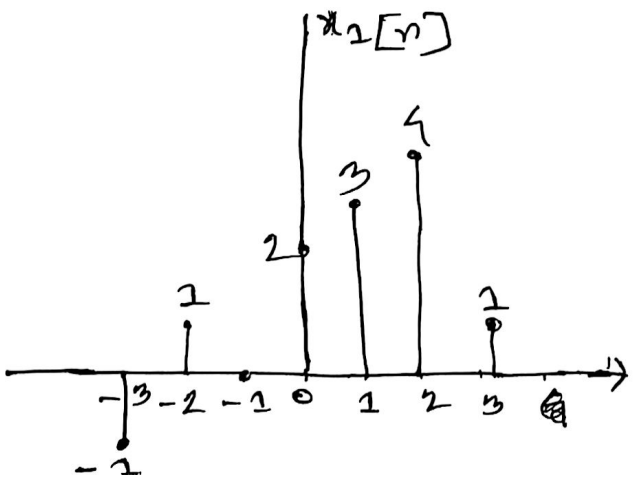



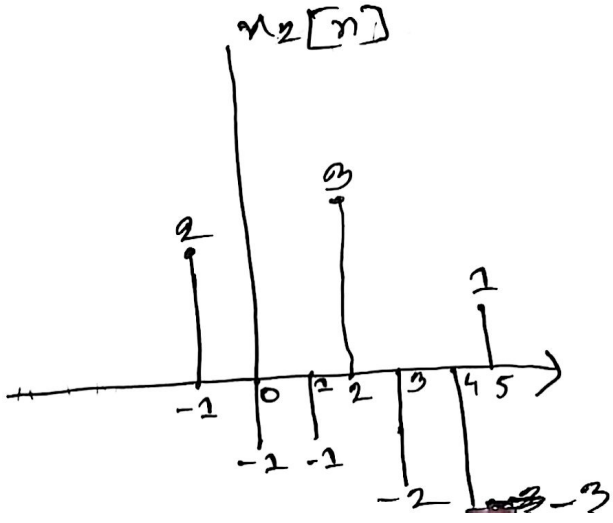

if,

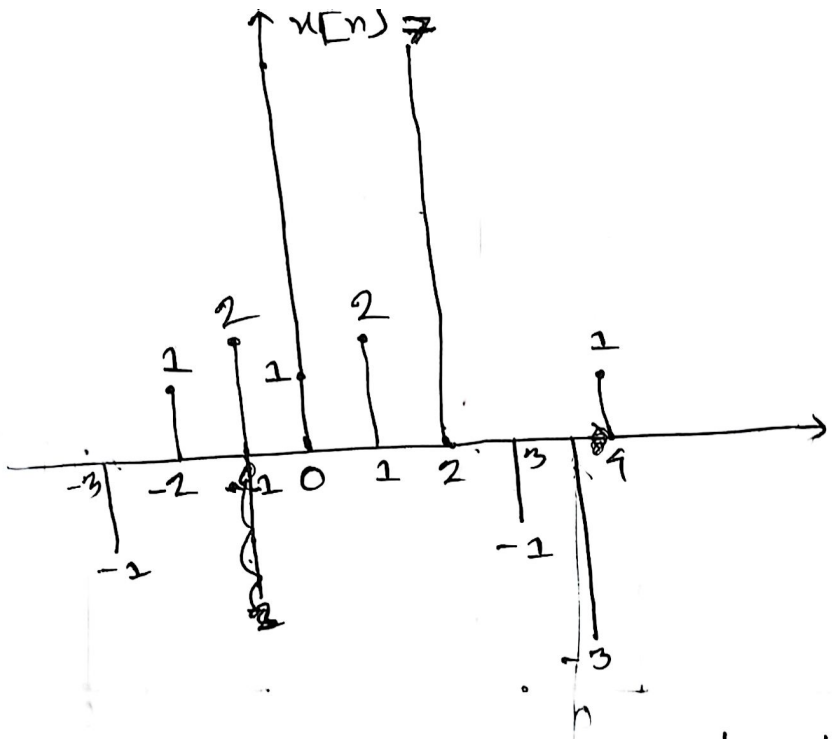
$$x_1[n] = \{-1, 1, \underset{\uparrow}{0}, 2, 3, 4, 1\}$$

$$x_2[n] = \{2, \underset{\uparrow}{-1}, -1, 3, -2, -3, 1\}$$

$$x[n] = \{-1, 1, 2, 2, 2, 7, -1, -3, 1\}$$





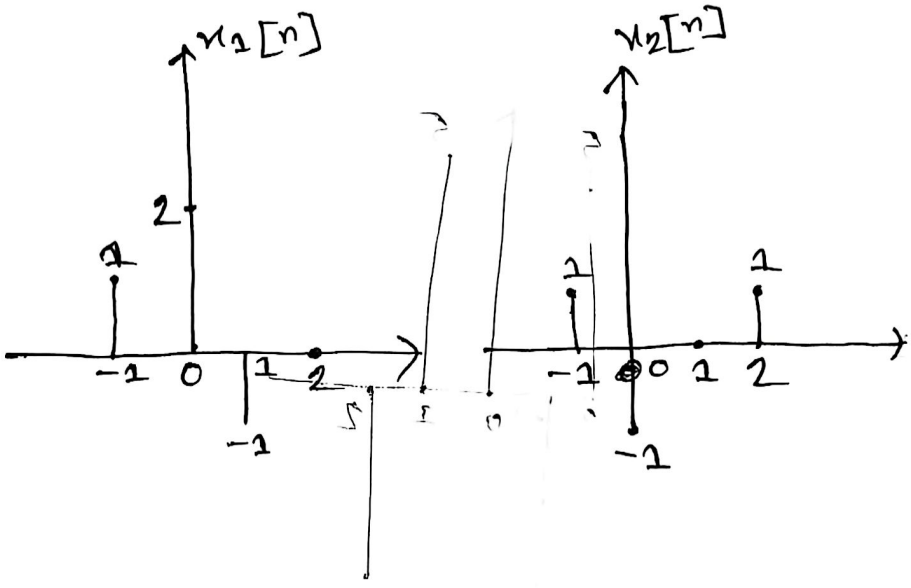

2. Multiplying: Signal Multiplication is also ~~known~~ as element-wise operation carried out at matching time indices n. Mathematically, $\boxed{x[n] = x_1[n] \cdot x_2[n]}$

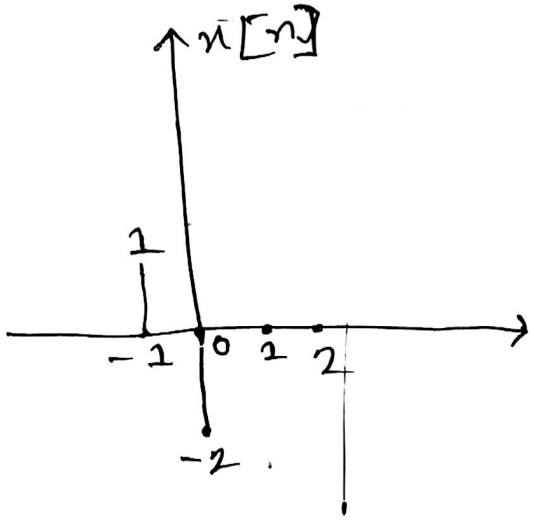
$x_1[n] = \{1, \underset{\uparrow}{2}, -1, 0\}$

$x_2[n] = \{\underset{\uparrow}{1}, -1, 0, 1\}$

$x[n] = x_1[n]\, x_2[n]$

$= \{1, \underset{\uparrow}{-2}, 0, 0\}$



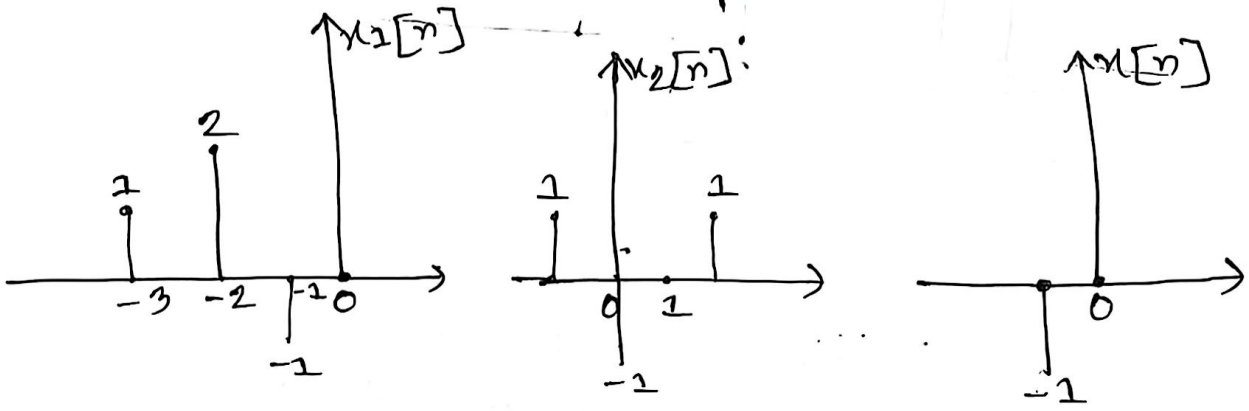



$x_1[n] = \{1, 2, -1, \underset{\uparrow}{0}\}$

$x_2[n] = \{1, \underset{\uparrow}{-1}, 0, 1\}$

$x[n] = x_1[n] \cdot x_2[n]$

$= \{-1, \underset{\uparrow}{0}\}$



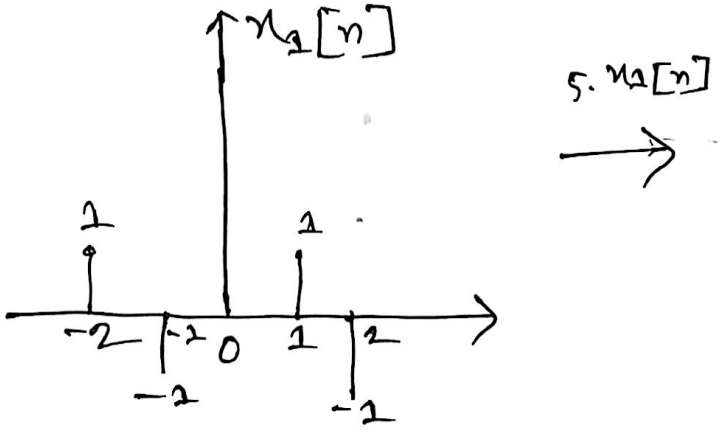

3. Amplitude Scaling: Amplitude scaling involves multiplying every sample of a single sequence by a constant scalar factor A. Mathematically:

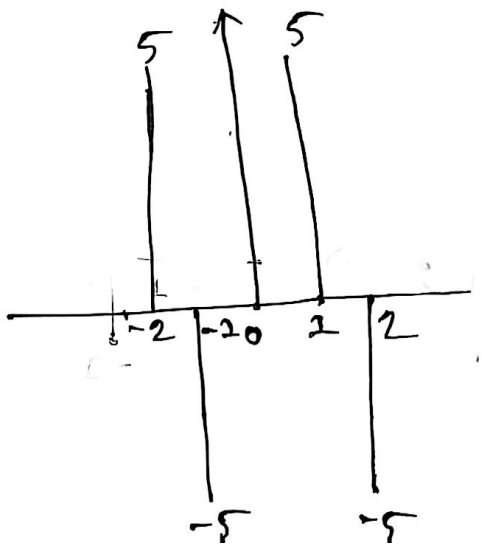
$$x_2[n] = A \cdot x_1[n]$$

এখানে value Gain হচ্ছে।

$x_1[n] = [1, -1, \underset{\uparrow}{0}, 1, -1]$

$x_2[n] = 5 \cdot x_1[n]$



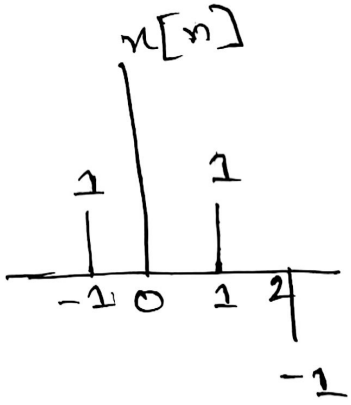
4. Time Scaling : Time Scaling modifies the time index n itself, changing the speed or density of the signal.

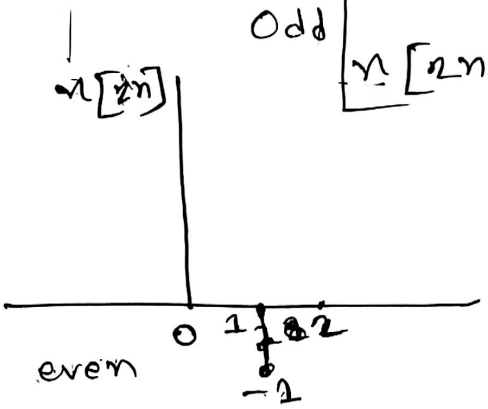
$x[n] = \{1, 0, 1, -1\}$ ↑

formulas:

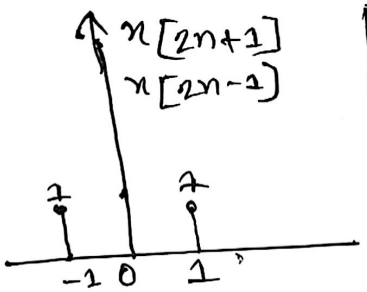
even $[x[2n]]$

Odd $\begin{cases} x[2n+1] & \text{if } n < 0 \\ x[2n-1] & \text{if } n > 0 \end{cases}$




$x[2n]$ ⇒ where $n = 0, 1, 2, -1$




for $[2n+1]$ $n = -1$

$2 \cdot -1 + 1 \Rightarrow x[-1] = 1$

for $[2n-1]$ $n = 1, 2$

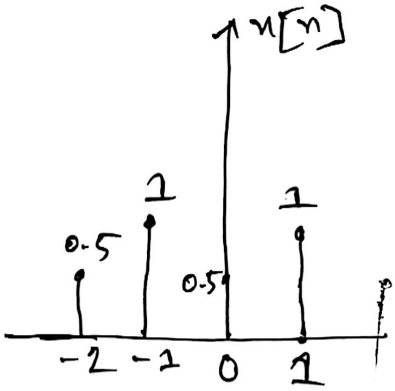
$2 \cdot 1 - 1 \Rightarrow x[1] = 1$

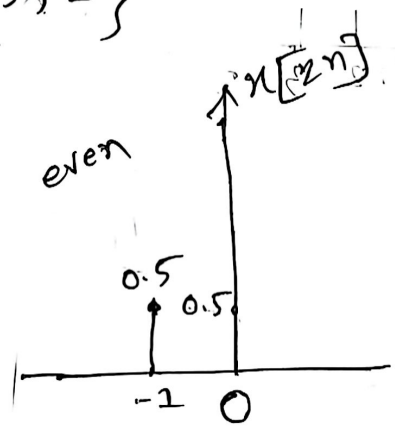
for understanding

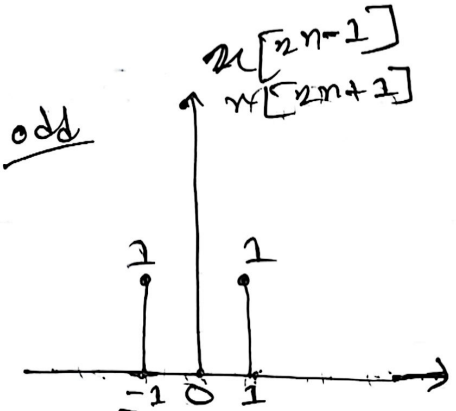
Another example:

$x[n] = \{0.5, 1, 0.5, 1\}$

$n = 0, 1, -1, -2$ ↑




$n = 0, 1, -1, -2$ in $x[2n]$

even

$n = 0 \rightarrow x[0]$

$n = 1 \rightarrow x[2]$ ✗

$n = -1 \rightarrow x[-2]$

$n = -2 \rightarrow x[-4]$ ✗

$x[2n-1]$

$n = 1 \Rightarrow x[1] = 1$

$x[2n+1]$

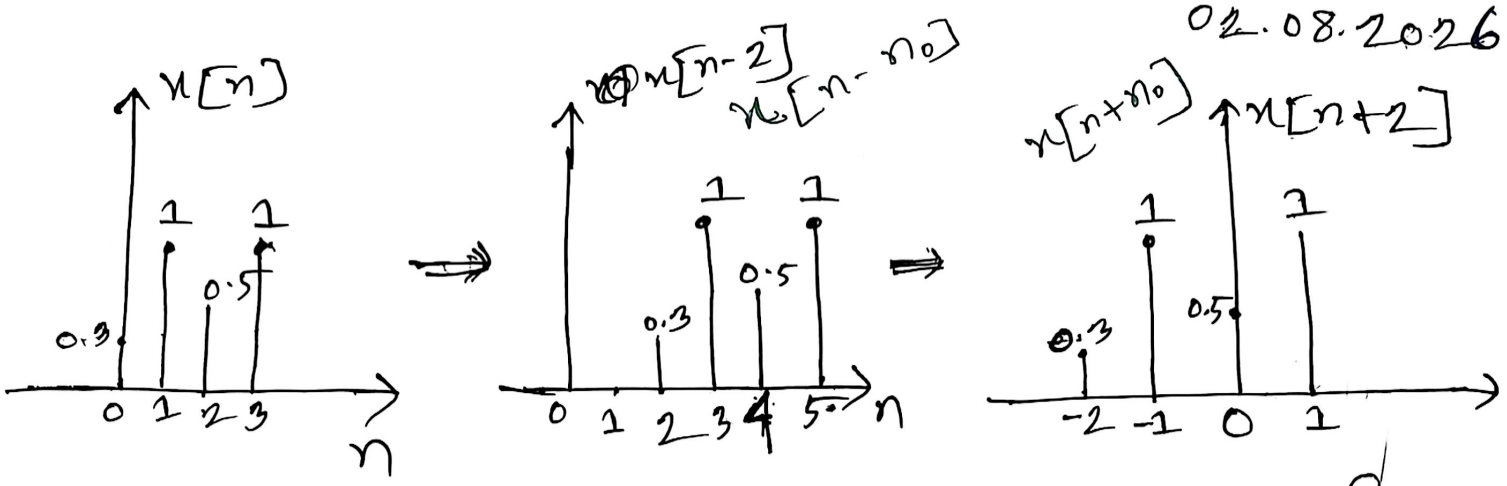
$n = -1 \Rightarrow x[-1] = 1$

$n = -2 \Rightarrow x[-3]$ ✗

# Time Shifting

Lecture-4

02.08.2026



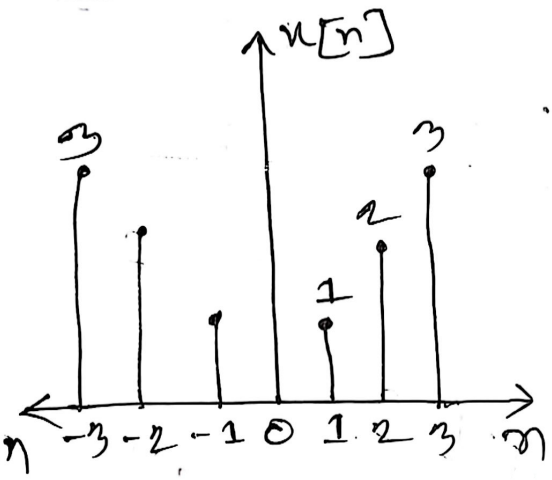

Main Signal  Delayed Signal  Advanced Signal

**Definition:** Time shifting modifies the independent time index $n$ of sequence $x[n]$ by an integer shift $n_0$.

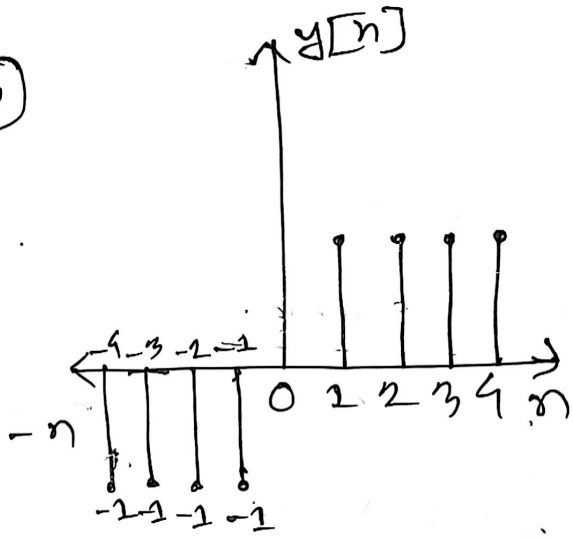
## Examples:

Ex.1.3

(a)




(b)




From (a) & (b) ↓

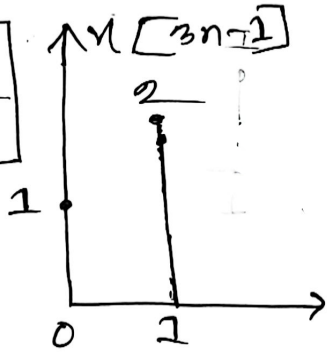
Questions:

(i) $x[2n]$

$n = -3$ to $n = 3$

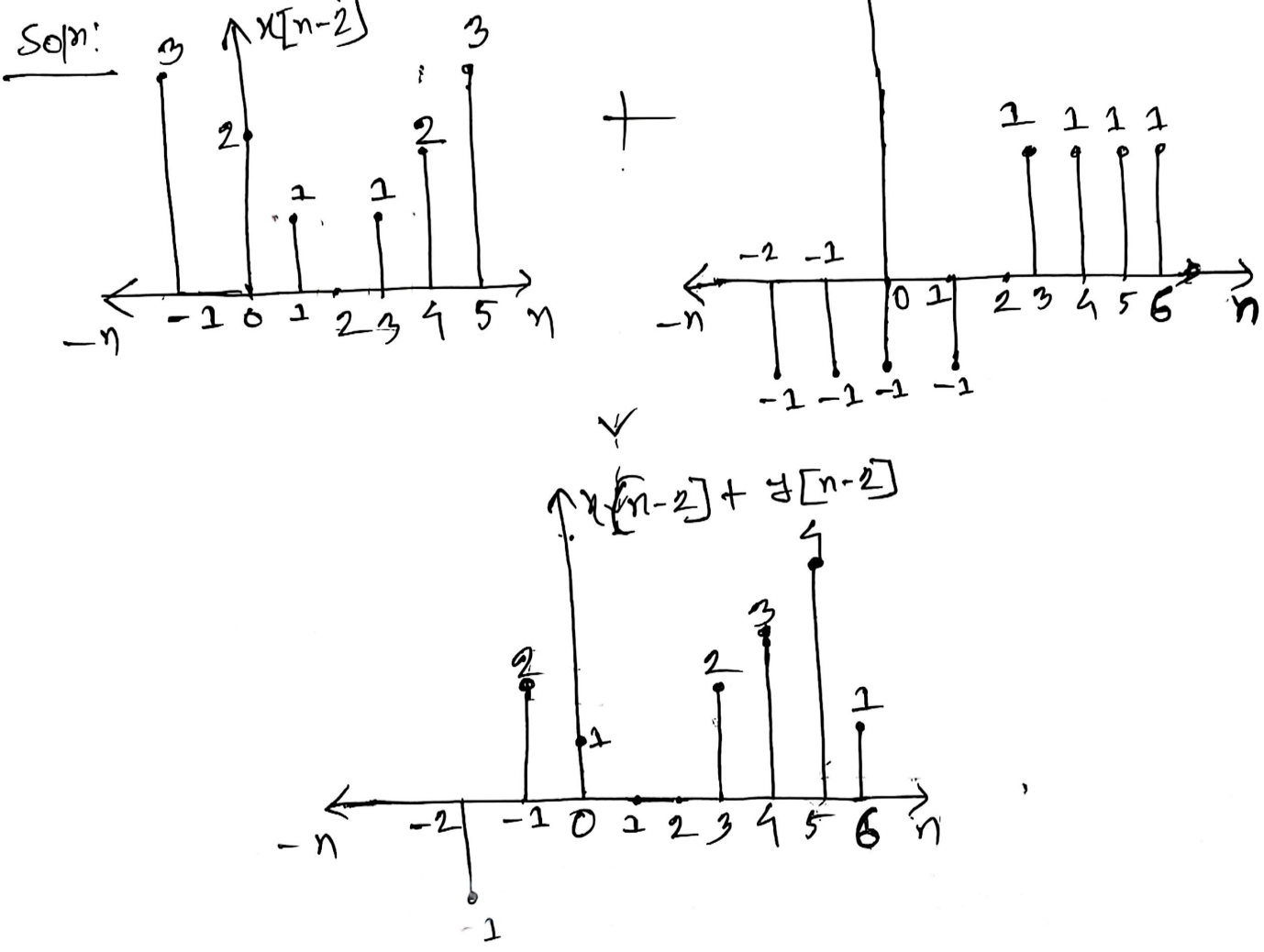
$n = -3$ to $n = 3$

(ii) $x[3n-1]$

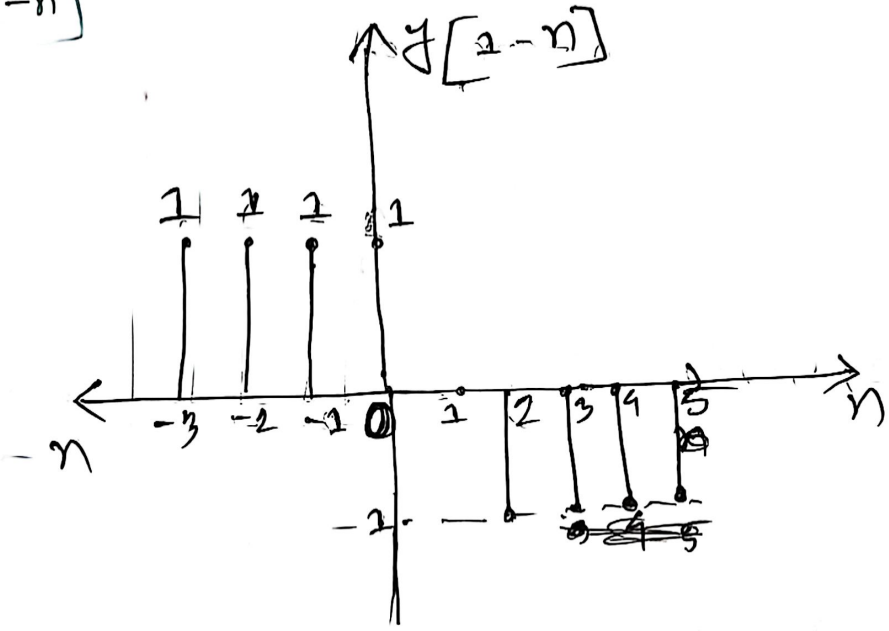


(iii) $x[n-2] + y[n-2]$

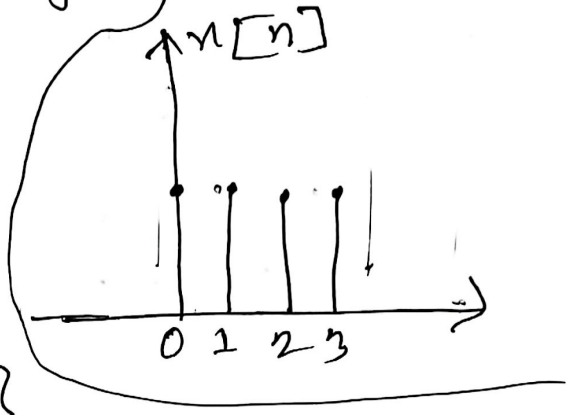
Soln:




(iv) $y[1-n]$



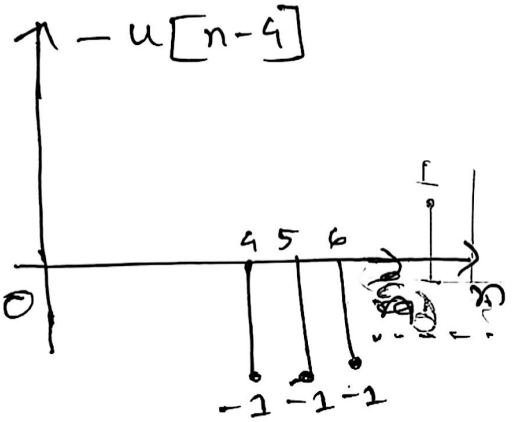
Ex.1.4 Consider the following signal and Represent the signal in terms of step signal.




Soln: Here,

$$x[n] = \{1, 1, 1, 1\}$$

$$u[n] = \{1, 1, 1, 1, 1, \ldots\}$$




$$x[n] = u[n] + \{-u[n-4]\}$$

$$\therefore [n] = u[n] - u[n-4]$$

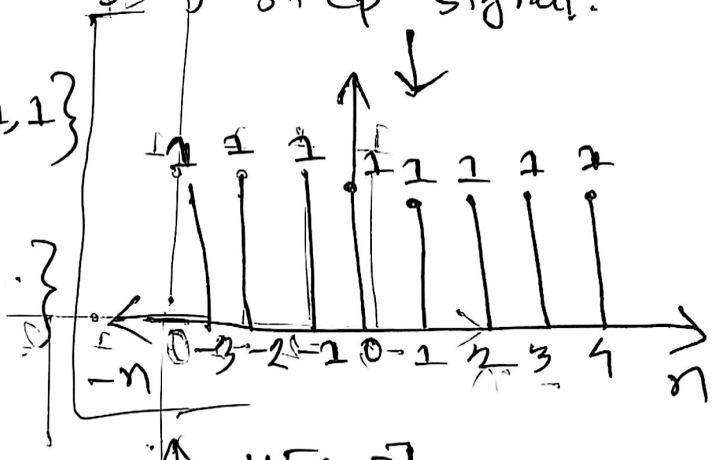
(Ans)

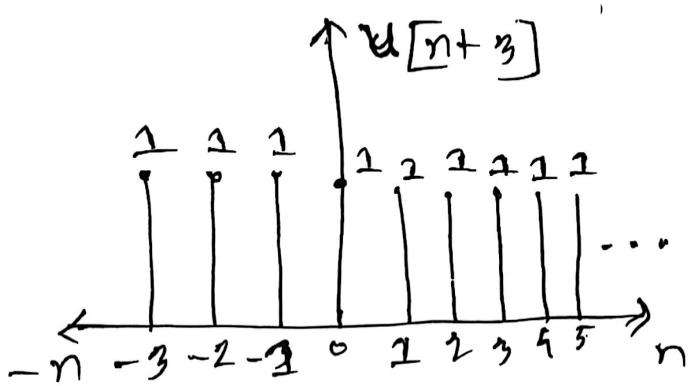
Example-1.5 ..... in terms of step signal.

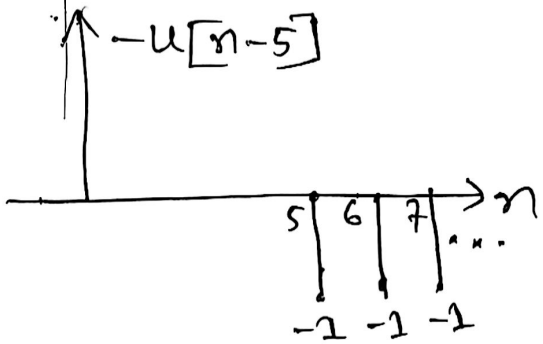
Soln: Here,

$$x[n] = \{1, 1, 1, 1, 1, 1, 1, 1\}$$

$$u[n] = \{1, 1, 1, 1, \ldots\}$$



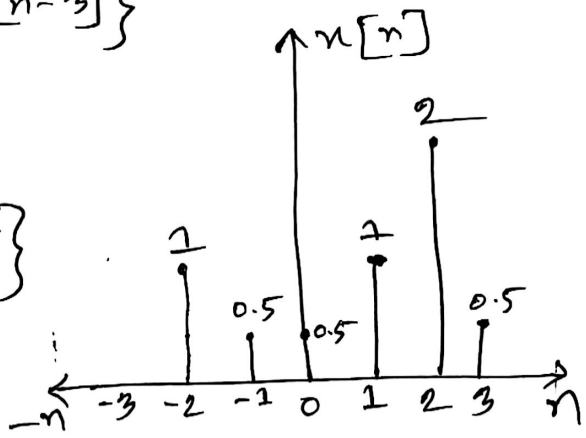



$$\therefore x[n] = u[n+3] + \{-u[n-5]\}$$

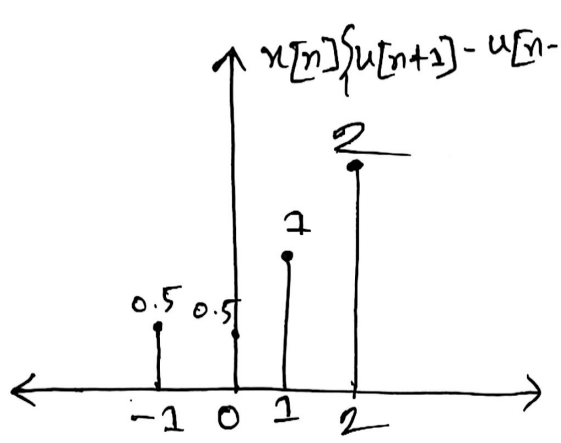
$$= u[n+3] - u[n-5] \quad \text{(Ans)}$$

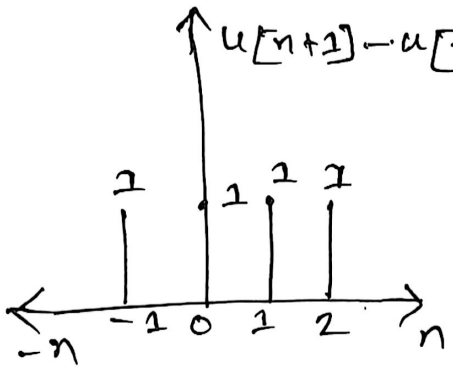
**Example-1.6** A discrete time sequence $x[n]$ is shown in below. Find $x[n]\{u[n+1] - u[n-3]\}$




**Soln** Here,

$$x[n] = \{1, 0.5, 0.5, 1, 2, 0.5\}$$



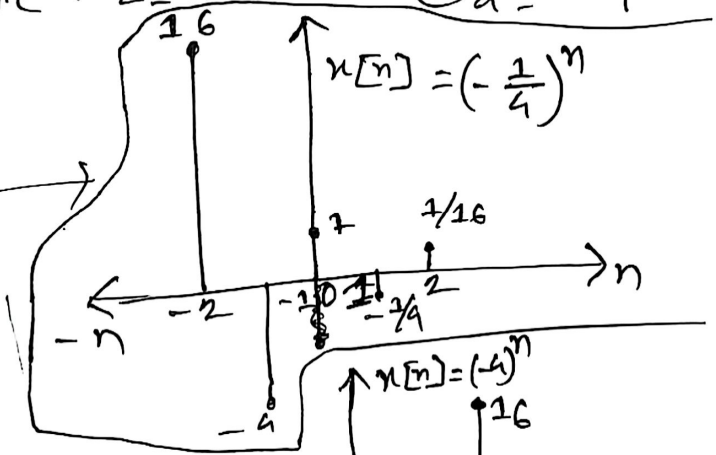



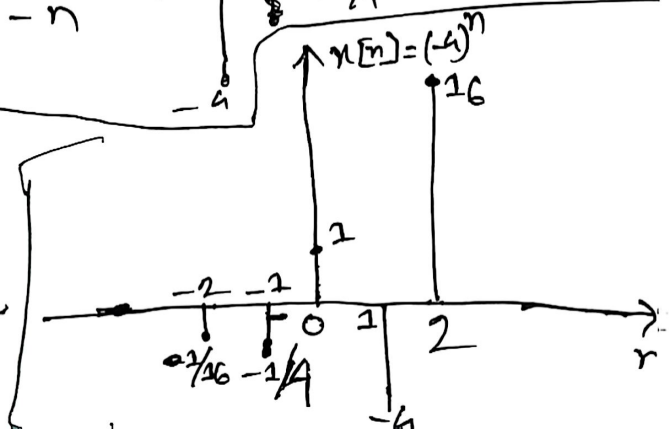

**Example-1.7** Sketch $x[n] = a^n$ where $-2 \le n \le 2$ for (i) $a = \left(-\frac{1}{4}\right)$ (ii) $a = -4$

**Soln:** for $x[n] = \left(-\frac{1}{4}\right)^n$

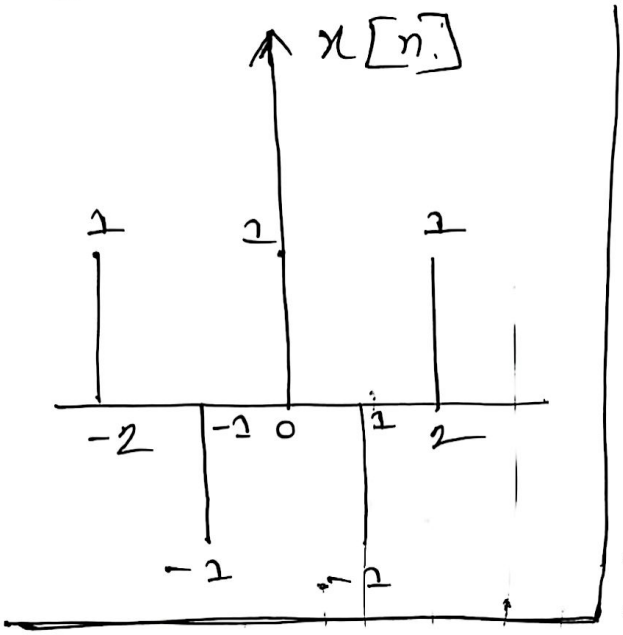
$$x[n] = (-4)^n$$

| n | -2 | -1 | 0 | 1 | 2 |
|---|---|---|---|---|---|
| $x[n] = \left(-\frac{1}{4}\right)^n$ | 16 | -4 | 1 | $-\frac{1}{4}$ | $\frac{1}{16}$ |
| $x[n] = (-4)^n$ | $\frac{1}{16}$ | $-\frac{1}{4}$ | 1 | -4 | 16 |

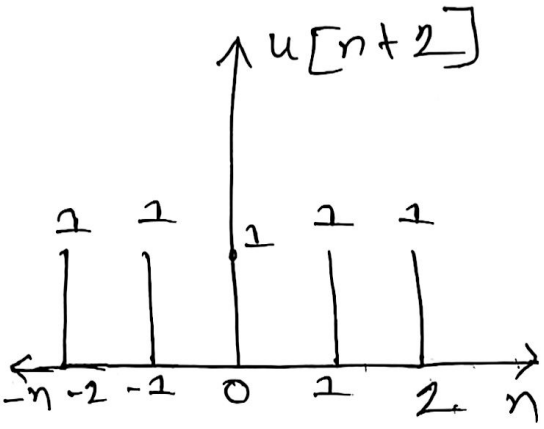


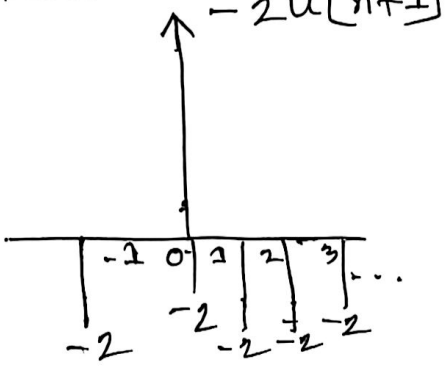
Example-1.8 Express $x[n] = (-1)^n$ as a sum of scaled and shifted step function, where $-2 \le n \le 2$



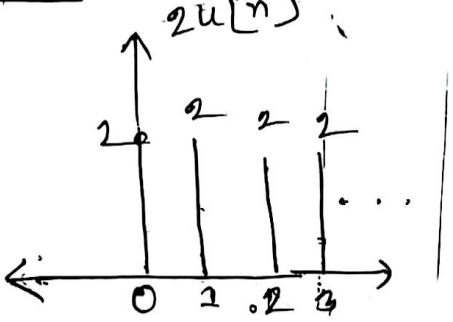

step-1



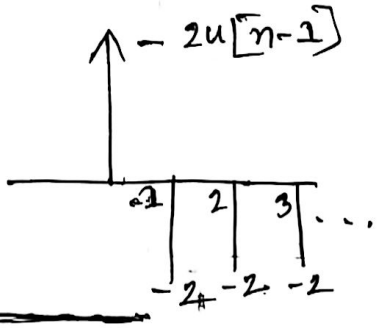

step-2



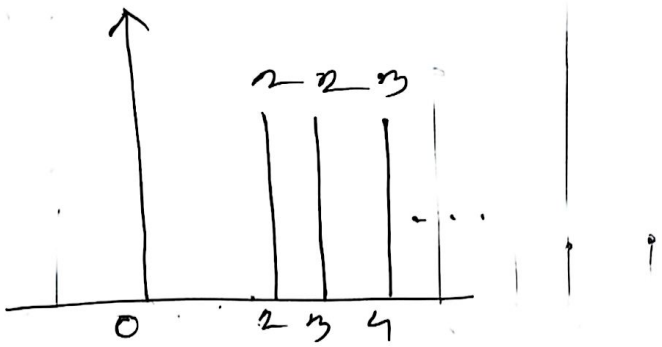

step-3




step-4

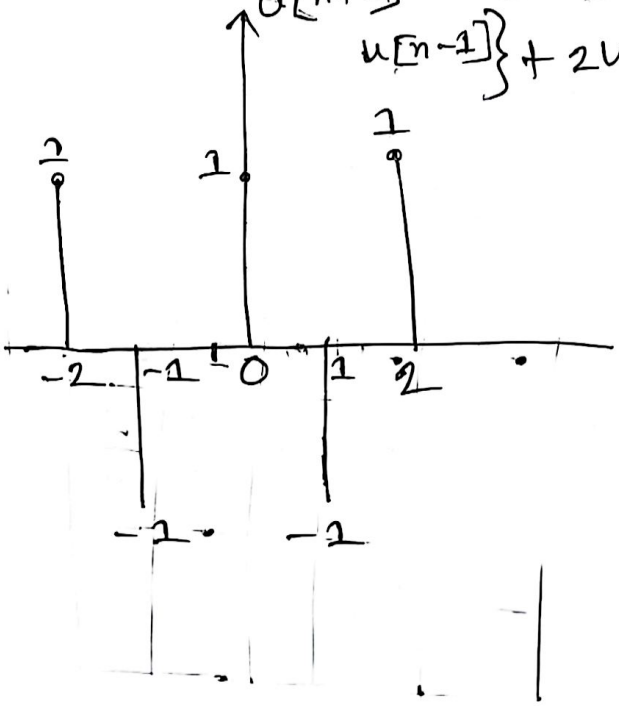



step-5



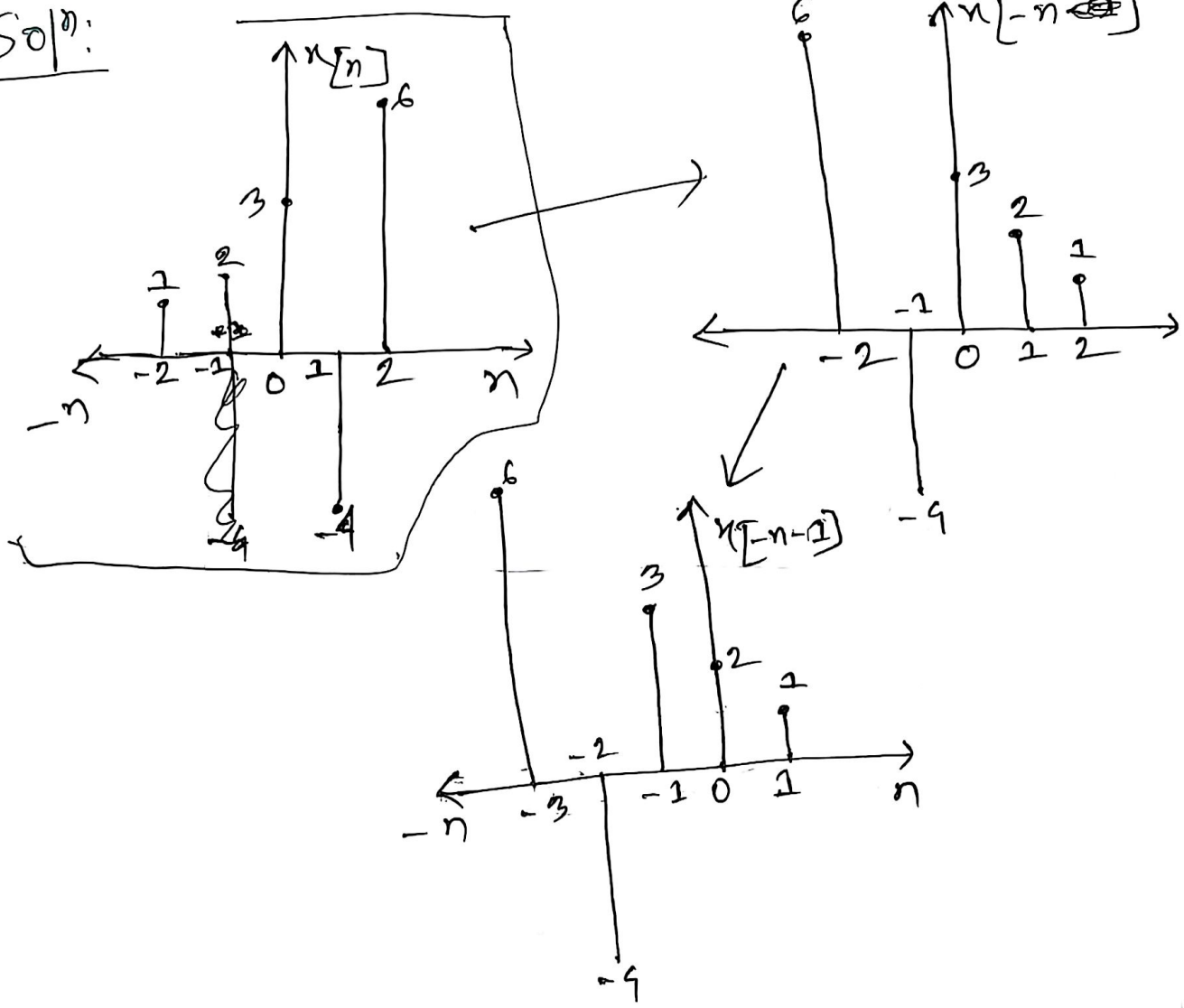

5-6

$u[n+2] - 2u[n+1] + 2\{u[n] - u[n-1]\} + 2u[n-2]$



Example-1.9 Given $x[n] = \{1, 2, 3, -4, 6\}$. Plot the signal

Soln:




Example-1.8 বাকি আছে.

$\therefore x[n] = u[n+2] - 2u[n+1] + 2\{u[n] - u[n-1]\} +$

$2u[n-2]$

# 1.6 Classification of Discrete-Time Signal

Lecture-5

09.08.2026

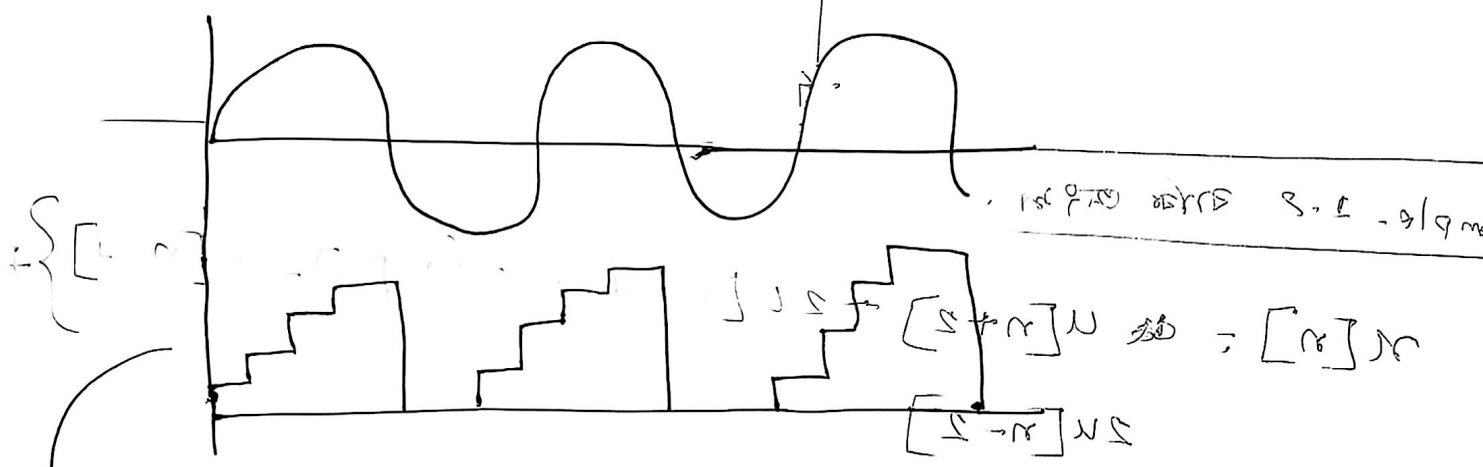
i) Periodic and Non-Periodic (aperiodic) Signal

ii) Odd and Even signal

iii) Energy and Power signal

(i) **Periodic and Non-Periodic DT signal:** A discrete time signal (sequence) $x[n]$ is said to be periodic with period N which is a positive integer if

$$x[n+N] = x[n] \quad \text{for all } n.$$



$$\sin\theta = \sin(2\pi+\theta)$$
$$= \sin(4\pi+\theta)$$
$$= \sin(20\cdot 2\pi+\theta)$$

**Properties:** Periodic Signal denoted as $x[n+N] = x[n]$

where N is samples/period/cycle.

$\sin(2\pi + \theta) = \sin\theta$

Here, periodicity $\longrightarrow 2\pi$

$\sin(2 \cdot 2\pi + \theta) = \sin\theta$

if $x[n] = \cos(\omega n + \theta)$ periodic or not?

**Ans:** $x[n] = \cos(\omega n + \theta)$

$$x[n+N] = \cos\{\omega(n+N) + \theta\}$$
$$= \cos\{\omega n + \omega N + \theta\}$$
$$= \cos(\omega N + \omega n + \theta)$$
$$= \cos(m 2\pi + \omega n + \theta)$$
$$= \cos(\omega n + \theta)$$

$\therefore x[n+N] = x[n]$

Here,

$\omega N = m 2\pi$

where, $m = 0, 1, 2, 3, \ldots$

and when period is fundamental period then $m = 1$

$N = \frac{2\pi}{\omega} m$

and $\omega N = m \cdot 2\pi$

$\Rightarrow 2\pi f \cdot N = m \cdot 2\pi \quad [\because \omega = 2\pi f]$

for fundamental period $m = 1$ then

$N = \frac{2\pi}{\omega}$

$\therefore f = \frac{m}{N}$ → always true for DT-signal

দুইটা পূর্ণসংখ্যার অনুপাত called rational number.

$$e^{i\theta} = \cos\theta + i\sin\theta$$

$$e^{i2\pi} = \cos(2\pi) + i\sin(2\pi) \qquad (\theta = 2\pi)$$

$$= 1 + i \cdot 0$$

$$= 1$$

formula:

$$e^{j\omega N}$$
$$= e^{j2\pi f N}$$
$$= e^{j2\pi \frac{m}{N} N}$$
$$= e^{j2\pi m}$$
$$= 1$$

---

for fundamental period

$m = 1$ then

$$N = \frac{2\pi}{\omega}$$

$$\left[ \omega = 2\pi f \right]$$

always the value is integer number.

$\frac{1}{N}$ ← called rational number for DT-signal

Lecture-6
12.08.2026

**Example-1.10:** Find the following complex signal is periodic or non-periodic and the fundamental period. $\longrightarrow x[n] = e^{j\omega n}$

**Ans:** Now, $x[n+N] = e^{j\omega(n+N)}$

$$= e^{j\omega n} \; e^{j\omega N}$$

$$= x[n] \cdot 1 \quad \left[\because e^{j\omega N} = 1\right]$$

$$= x[n]$$

$\therefore x[n+N] = x[n]$ then the signal is periodic.

**Example-1.11:** $x[n] = \sin(\omega n + \phi)$. under which condition, above signal is periodic. $\longrightarrow x[n] = \sin(\omega n + \phi)$

**Soln:** Now, $x[n] = \sin(\omega n + \phi)$

$$x[n+N] = \sin(\omega(n+N) + \phi)$$

$$= \sin(\omega n + \omega N + \phi)$$

$$= \sin(2\pi f N + \omega n + \phi)$$

$$= \sin\left(2\pi \frac{m}{N} N + \omega n + \phi\right)$$

$$= \sin(m \cdot 2\pi + \omega n + \phi)$$

$$= \sin(\omega n + \phi)$$

$$= x[n]$$

$\therefore x[n+N] = x[n]$ $\therefore x[n]$ signal is periodic.

$$\cos(A+B) = \cos A \cos B - \sin A \sin B$$

$$\therefore \cos(\theta+\theta) = \cos\theta\cos\theta - \sin\theta\sin\theta \quad [A = B = \theta]$$

$$\boxed{\cos 2\theta = \cos^2\theta - \sin^2\theta}$$

$$\cos 2\theta = 1 - \sin^2\theta - \sin^2\theta$$

$$\Rightarrow \cos 2\theta = 1 - 2\sin^2\theta$$

$$\Rightarrow 2\sin^2\theta = 1 - \cos 2\theta$$

$$\therefore \sin^2\theta = \frac{1}{2} - \frac{1}{2}\cos 2\theta \Big\} \text{power reduction identity}$$

square বাকলে দুইটা মান হবে।

$$x[n] = \sin^2(\omega n + \theta) \checkmark$$

$$x[n] = \frac{1}{2} - \frac{1}{2}\cos(2\omega n + 2\theta) \checkmark$$

Example-1.12 → follow book.

Example-1.13: Find whether the following signals are periodic. If periodic, determine the fundamental period.

a) $x[n] = e^{j\pi n}$

b) $x[n] = \cos\left[\frac{n}{8} - \pi\right]$

c) $x[n] = \sin^2\frac{\pi}{4}n$

Soln:

(a) $x[n] = e^{j\pi n}$

Here, $\omega = \pi$

$$N = \frac{2\pi}{\omega}m \Rightarrow N = \frac{2\pi}{\pi}; \text{ for } m = 1$$

$$N = 2$$

$x[n]$ is periodic with fundamental period 2.

(b) Soln: $x[n] = \cos\left[\frac{n}{8} - \pi\right]$

Here, $\omega = \frac{1}{8}$

$$N = \frac{2\pi}{\omega}m = 16\pi m$$

For any integer value of $m$, $N$ is not integer.

Hence $x[n]$ is no periodic

(c) $x[n] = \sin^2 \frac{\pi}{4} n$

$= \frac{1}{2} - \frac{1}{2} \cos \frac{2\pi}{4_2} n$

$= x_1[n] + x_2[n]$

$x_1[n] = \frac{1}{2} = \frac{1}{2}(1)^n$ is periodic with $N_1 = 1$

$x_2[n] = -\frac{1}{2} \cos \frac{\pi}{2} n$ ; Here, $\omega = \frac{\pi}{2}$

$\therefore N_2 = \frac{2\pi}{\omega} m = 4$; for $m=1$ & $\omega = \frac{\pi}{2}$

$\therefore \frac{N_1}{N_2} = \frac{1}{4} \Rightarrow 4N_2 = N_1 = N$

$\therefore$ Fundamental Period is $N = 4$

Example-1.14 Find the periodicity of the following DT signal: $x[n] = \sin \frac{2\pi}{3} n + \cos \frac{\pi}{2} n$

Soln: $x[n] = \sin \frac{2\pi}{3} n + \cos \frac{\pi}{2} n$

$= x_1[n] + x_2[n]$

Now, $x_1[n] = \sin \frac{2}{3} \pi n$ , $N = \frac{2\pi}{\omega_1} m_1 = \frac{2\pi}{2\pi} 3 m_1$

$\omega_1 = \frac{2}{3} \pi$ $= 3$ [for $m_1 = 1$]

again, $x_2[n] = \cos\frac{\pi}{2}n$

$\omega_2 = \frac{\pi}{2}$ ; $N_2 = \frac{2\pi}{\omega} m_2 = \frac{2\pi}{\pi} 2m_2$

$= 4$ [for $m_2 = 1$]

$\frac{N_1}{N_2} = \frac{3}{4} \Rightarrow 4N_1 = 3N_2 = N$

$N = 12$

∴ Periodicity is $N = 12$

Example:1.15 : $x[n] = \cos\left(\frac{n\pi}{2}\right)\cos\left(\frac{n\pi}{4}\right)$

Soln:

$x[n] = \cos\left(\frac{n\pi}{2}\right)\cos\left(\frac{n\pi}{4}\right)$

Now,

$x_1[n] = \cos\frac{n\pi}{2}$ ∴ $\omega_1 = \frac{\pi}{2}$

∴ $N = \frac{2\pi}{\omega_1} m_1 = \frac{2\pi}{\pi} 2m_1 = 4$ [for $m_1 = 1$]

$x_2[n] = \cos n\frac{\pi}{4}$ ; $\omega_2 = \frac{\pi}{4}$

∴ $N = \frac{2\pi}{\omega_2} m_2 = \frac{2\pi}{\pi} 4 m_2 = 8$ [for $m_2 = 1$]

$\frac{N_1}{N_2} = \frac{4}{8} = \frac{1}{2}$ ∴ $N = 8$

$\Rightarrow 2N_2 = N_1 = N$

Sir's method:

We Know, $\cos(A)\cos(B) = \frac{1}{2}[\cos(A+B) + \cos(A-B)]$

Applying this to $x[n]$:

$$x[n] = \frac{1}{2}\left[\cos\left(\frac{n\pi}{2}+\frac{n\pi}{4}\right) + \cos\left(\frac{n\pi}{2}-\frac{n\pi}{4}\right)\right]$$

$$x[n] = \frac{1}{2}\cos\left(\frac{3\pi}{4}n\right) + \frac{1}{2}\cos\left(\frac{\pi}{4}n\right)$$

$\therefore x[n] = x_1[n] + x_2[n]$

Here,
$x_1[n] = \frac{1}{2}\cos\left(\frac{3\pi}{4}n\right)$
$x_2[n] = \frac{1}{2}\cos\left(\frac{\pi}{4}n\right)$

for $x_2[n]$

$$x_1[n+N_1] = \frac{1}{2}\cos\left(\frac{3\pi}{4}(n+N_1)\right)$$

$$= \frac{1}{2}\cos\left(\frac{3\pi}{4}n + \frac{3\pi}{4}N_1\right)$$

For $x_1[n+N_1] = x_1[n]$ phase shift must be integer multiple of $2\pi$.

$$\frac{3\pi}{4}N_1 = 2\pi m_1$$

$$\Rightarrow N_1 = \frac{8}{3}\times 3 \quad [m_1 = 3 \text{ to make } N_1 \text{ integer}]$$

$\therefore N_1 = 8$

again,

$$x_2[n+N_2] = \frac{1}{2}\cos\left(\frac{\pi}{4}(n+N_2)\right)$$

$$= \frac{1}{2}\cos\left(\frac{\pi}{4}n + \frac{\pi}{4}N_2\right)$$

for $x_2[n+N_2] = x_2[n]$

$$\frac{\pi}{4} N_2 = 2\pi m_2$$

$$\Rightarrow N_2 = 8m_2$$

$$\therefore N_2 = 8 \quad [\text{for } m_2 = 1]$$

$\therefore$ Fundamental period of $x[n]$ is $N = 8$

Example-1.16 periodic or not? fundamental period...

Soln: (a) $x[n] = \underbrace{\cos\left(\frac{\pi}{2}n\right)}_{\text{say: } x_1[n]} + \underbrace{\sin\left(\frac{\pi}{8}n\right)}_{x_2[n]} + \underbrace{3\cos\left(\frac{\pi}{4}n + \frac{\pi}{3}\right)}_{x_3[n]}$

Now, $x[n+N_1] = \cos\left(\frac{\pi}{2}(n+N_1)\right) = \cos\left(\frac{\pi}{2}n + \frac{\pi}{2}N_1\right)$

for $x_1[n+N_1] = x_1[n]$;

$$\frac{\pi}{2} N_1 = 2\pi m_1$$

$$\therefore N_1 = 4 \quad [\text{for } m_1 = 1]$$

again,

$$x[n+N_2] = \sin\left(\frac{\pi}{8}(n+N_2)\right) = \sin\left(\frac{\pi}{8}n + \frac{\pi}{8}N_2\right)$$

for, $x_1[n+N_2] = x_2[n]$;

$$\frac{\pi}{8} N_2 = 2\pi m_2$$

$$\therefore N_2 = 16 \quad [\text{for } m_2 = 1]$$

then,

$$x_3[n+N_3] = 3\cos\left(\frac{\pi}{4}(n+N_3)+\frac{\pi}{3}\right) = 3\cos\left(\frac{\pi}{4}n+\frac{\pi}{3}+\frac{\pi}{4}\right.$$

for $x_3[n+N_3] = x_3[n]$;

$$\frac{\pi}{4}N_3 = 2\pi m_3$$

$$\therefore N_3 = 8 \quad [\text{for } m_3 = 1]$$

$$\therefore N = LCM(N_1, N_2, N_3) = LCM(4, 16, 8) = 16$$

$\therefore$ Periodicity $N = 16$ and $x[n]$ is periodic.

(b) $$x[n] = \underbrace{e^{j\frac{2\pi}{3}n}}_{x_1[n]} + \underbrace{e^{j\frac{3\pi}{4}n}}_{x_2[n]}$$

Now, $$x_1[n+N_1] = e^{j\frac{2\pi}{3}(n+N_1)} = e^{j\frac{2\pi}{3}n} \cdot e^{j\frac{2\pi}{3}N_1}$$

for $x_1[n+N_1] = x_1[n]$, we require:

$$e^{j\frac{2\pi}{3}N_1} = 1$$

$$\therefore \frac{2\pi}{3}N_1 = 2\pi m_1$$

$$\therefore N_1 = 3 \quad [\text{for } m_1 = 1]$$

again, $$x_2[n+N_2] = e^{j\frac{3\pi}{4}(n+N_2)} = e^{j\frac{3\pi}{4}n} \cdot e^{j\frac{3\pi}{4}N_2}$$

for $x_2[n+N_2] = x_2[n]$, we require;

$e^{j\frac{3\pi}{4}N_2} = 1 \quad \therefore \quad \frac{3\pi}{4}N_2 = 2\pi m_2$

$\therefore \quad N_2 = 8 \quad [\text{for } m_2 = 3]$

$\therefore \quad N = LCM(N_1, N_2) = LCM(3, 8) = 24$

$\therefore$ $x[n]$ is periodic and fundamental period $N = 24$.

---

## 1.6.2 Odd and Even functions

Lecture-7
19.08.2020

$\sin\theta \rightarrow$ Odd
$\cos\theta \rightarrow$ Even

| Odd | Even |
|---|---|
| $\sin(-\theta) = -\sin\theta$ | $\cos(-\theta) = \cos\theta$ |
| $x[-n] = -x[n]$ | $x[-n] = x[n]$ |

formula, $x[n] = x_e[n] + x_o[n]$

where, $x_e[n] = \frac{1}{2}\{x[n] + x[-n]\}$

$x_o[n] = \frac{1}{2}\{x[n] - x[-n]\}$

proof:

$x[n] = x_e[n] + x_o[n]$

$= \frac{1}{2}\{x[n] + x[-n]\} + \frac{1}{2}\{x[n] - x[-n]\}$

$= \frac{1}{2}\{x[n] + x[-n] + x[n] - x[-n]\}$

$= \frac{1}{2}$

$$= \frac{1}{2}\{2x[n]\}$$

$$= x[n]$$

= Answer

1.18

Example-1.~~19~~: Consider $x[n] = \{-2, 1, \underset{\uparrow}{3}, -5, 4\}$

Determine the even and odd component.

Soln: Given, $x[n] = \{-2, 1, \underset{\uparrow}{3}, -5, 4\}$

$$x[-n] = \{4, -5, \underset{\uparrow}{3}, 1, -2\}$$

Now,

$$x_e[n] = \frac{1}{2}\{x[n] + x[-n]\}$$

$$= \frac{1}{2}\{(-2+4), (1-5), \underset{\uparrow}{(3+3)}, (-5+1), (4-2)\}$$

$$= \frac{1}{2}\{2, -4, \underset{\uparrow}{6}, -4, 2\}$$

$$= \{1, -2, 3, -2, 1\}$$

$$x_o[n] = \frac{1}{2}\{x[n] - x[-n]\}$$

$$= \frac{1}{2}\{(-2-4), (1+5), \underset{\uparrow}{(3-3)}, (-5-1), (4+2)\}$$

$$= \frac{1}{2}\{-6, 6, \underset{\uparrow}{0}, -6, 6\}$$

$$= \{-3, 3, \underset{\uparrow}{0}, -3, 3\}$$

$$\therefore x_e[n] + x_o[n] = x[n]$$

Example-1.17: Find whether the below functions are odd or even:

i) $x[n] = \sin 2\pi n$

ii) $x[n] = \cos 2\pi n$

Soln:

i) $x[n] = \sin 2\pi n$

$\Rightarrow x[-n] = \sin(-2\pi n) = -\sin 2\pi n$

$\therefore x[-n] = -x[n]$ ; This is an odd signal.

ii) $x[n] = \cos 2\pi n$

$\Rightarrow x[-n] = \cos(-2\pi n) = \cos 2\pi n$

$\therefore x[-n] = x[n]$ ; This is an even signal.

Example - 1.19: Find the even and odd components of following DT signal and sketch them.

$$x[n] = \{-2, 1, 2, -1, 3\}$$

Soln: Given, $x[n] = \{-2, 1, 2, -1, 3\}$

$$x[-n] = \{3, -1, 2, 1, -2\}$$

Now,

$$x_e[n] = \frac{1}{2}\{x[n] + x[-n]\}$$

$$= \frac{1}{2}\{(-2+3), (1-1), (2+2), (-1+1), (3-2)\}$$

$$= \frac{1}{2}\{1, 0, 4, 0, 1\}$$

$$= \{\frac{1}{2}, 0, 2, 0, \frac{1}{2}\}$$

$$x_o[n] = \frac{1}{2}\{x[n] - x[-n]\}$$

$$= \frac{1}{2}\{(-2-3), (1+1), (2-2), (-1,-1), (3+2)\}$$

$$= \frac{1}{2}\{-5, 2, 0, -2, 5\}$$

$$= \{-\frac{5}{2}, 1, 0, -1, \frac{5}{2}\}$$

$\therefore x_e[n] + x_o[n] = x[n]$

Lecture-8

23.08.2026

## 1.6.3 Energy & Power Signal

for a discrete time signal $x[n]$, the total energy is defined as $\boxed{E = \sum_{n=-\infty}^{\infty} |x(n)|^2}$

and the average power is defined as:

$$P = \lim_{N\to\infty} \frac{1}{(2N+1)} \sum_{n=-N}^{N} |x(n)|^2$$

From the definitions;

i) $x[n]$ is an energy sequence if $0 < E < \infty$
then average power $P = 0$

for $0 < E < \infty$ ; $P = 0$

ii) $x[n]$ is a power signal if $0 < P < \infty$
then, total Energy $E = \infty$

**Basic Example:** Consider the following signal,

$$u[n] = \begin{cases} 1, & n \geq 0 \\ 0, & n < 0 \end{cases}$$

find whether the signal is Energy or Power Signal

$$E = \sum_{n=-\infty}^{\infty} |u[n]|^2$$

$$= \sum_{-\infty}^{-1} |u[n]|^2 + \sum_{n=0}^{\infty} |u[n]|^2$$

$$= 0 + \{1^2 + 1^2 + 1^2 + \cdots \infty\}$$

$$= 1 + 1 + 1 + 1 + 1 + \cdots$$

$$= \infty$$

$$P = \lim_{N \to \infty} \frac{1}{2N+1} \sum_{n=-N}^{N} |u[n]|^2$$

$$= \lim_{N \to \infty} \frac{1}{2N+1} \sum_{n=-N}^{-1} |u[n]|^2 + \lim_{N \to \infty} \frac{1}{2N+1} \sum_{n=0}^{N} |u[n]|^2$$

$$= 0 + \lim_{N \to \infty} \frac{1}{2N+1} \{1^2 + 1^2 + 1^2 + 1^2 + \cdots + 1^2\}$$

$$= \lim_{N \to \infty} \frac{1}{2N+1} \{1 + 1 + 1 + 1 + \cdots + N\}$$

$$= \lim_{N\to\infty} \frac{1}{2N+1}(N+1)$$

$$= \lim_{N\to\infty} \frac{\frac{N+1}{N}}{\frac{2N+1}{N}}$$

$$= \lim_{N\to\infty} \frac{1+\frac{1}{N}}{2+\frac{1}{N}}$$

$N\to\infty$, $\frac{1}{N}\to\frac{1}{\infty}=0$

$$= \frac{1+0}{2+0} = \frac{1}{2}$$

$$\therefore P = \frac{1}{2}$$

$\therefore$ This is Power signal.

Example : Consider the following signal and find whether the signal is energy signal or power signal

$$\delta[n] = \begin{cases} 1, & n=0 \\ 0, & n\neq 0 \end{cases}$$

Sol$^{n}$:

$$E = \sum_{n=-\infty}^{\infty} |\delta[n]|^2$$

$$= \sum_{n=-\infty}^{-1} |\delta[n]|^2 + \sum_{n=0} |\delta[n]|^2 + \sum_{n=1}^{\infty} |\delta[n]|^2$$

$$= 0 + 1 + 0$$

$$= 1$$

Lecture-9

30.08.2024

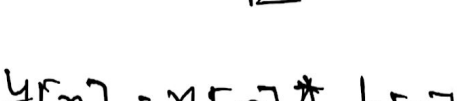
## Convolution

$x(n) \rightarrow$ input

$y(n) \rightarrow$ output

$y[n] = x[n] * h[n]$

$h[n] \rightarrow$ impulse response




$$y[n] = x[n] * h[n]$$

$$= \sum_{K=-\infty}^{\infty} x[K] \cdot h[n-k]$$

## Difference Eqn

present output past output এর উপর dependent

$$y[n] = k[n] + 0.5y[n-1]$$

↓ past

$$y[n] - 0.5y[n-1] = x[n]$$

↓ present

recursive system

## Correlation

Similarities bet$^n$ signals

→ convolution এর eqn

$$y[n] = \sum_{M=-\infty}^{\infty} x[n]\, h[n+m]$$

$$R_{xy}[m] = \sum_{M=-\infty}^{\infty} x[n]\, y[n+$$

---

**Correlation:** $R_{xy}[M] = \sum_{M=-\infty}^{\infty} x[n]\, y[n+m]$

$$R_{xy}[M] = \left\{ \begin{matrix} 4, 5, \underline{1}0, 3, 2 \\ -2, -1, 0, 1, 2 \end{matrix} \right\}$$

2টা value different হলে মানে peak এর মান অনেক difference থাকলে → Cross correlation

2টা same type signal এর মধ্যে correlation হলেই তার 'same' মানে - even(signal) correlation.

minus/plus এর দিক যেমন হবে

$$\begin{Bmatrix} 1 & 2 & 10 & 2 & 1 \\ -2 & -1 & 0 & 1 & -1 \end{Bmatrix}$$

{ 2টি same signal correlation করলে → auto correlation

auto-correlation করলে 0 তে peak পাবো even signal পাবো - যত সময় নেবে তত time period.

correlation two types → cross-correlation
→ Auto → even signal পাই।

Example-1.48 Determine the output of the recursive system described by the following differeace eqn: $y[n+2] + 3y[n+1] + 2y[n] = u[n]/\delta[n]$

where, $y[0] = 0$, $y[1] = 1$

$$u[n] = \begin{cases} 1, & n \geq 0 \\ 0, & n < 0 \end{cases}$$

$$\delta[n] = \begin{cases} 1, & n = 0 \\ 0, & n \neq 0 \end{cases}$$

Sol$^n$: when, $n = 0$,

$$y[0+2] + 3y[0+1] + 2y[0] = u[0]$$

$$y[2] + 3y[1] + 2y[0] = u[0]$$

$$y[2] + 3.1 + 2.0 = u[0]$$

$$y[2] = 1 - 3 = -2$$

When, $n=1$,

$$y[1+2] + 3y[1+2] + 2y[1] = u[1]$$

$$y[3] + 3y[2] + 2y[1] = u[1]$$

$$y[3] + 3 \cdot -2 + 2 \cdot 1 = u[1]$$

$$y[3] - 6 + 2 = 1$$

$$y[3] = 1 + 6 - 2 = 5$$

When, $n=2$,

$$y[2+2] + 3y[2+1] + 2y[2] = u[2]$$

$$y[4] + 3y[3] + 2y[2] = 1$$

$$y[4] + 3 \cdot 5 + 2 \cdot -2 = 1$$

$$\therefore y[4] = -15 + 4 + 1 = -10$$

$$\therefore y[n] = \{0, 1, -2, 5, -10, \cdots\}$$

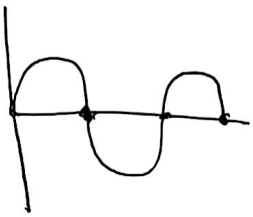
---

## Sampling

Lecture-10

06.09.2026

কোনো signal-কে computer এ নিতে হবে।



infinity data নিতে time লাগবে
data নিব।

এটা খুবের চায় recover করতে হবে।

recover হয়নি, হারিয়ে গেছে।

কারণ একটা specific frequency ensure করতে হবে।

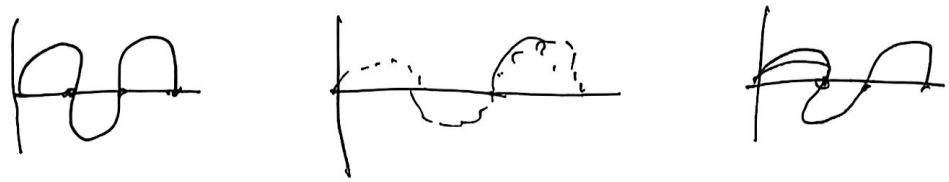
↳ ~~Nugui~~ Nyquist Theorem

$$f_s \geq 2 \cdot f_{max}$$

$f_s \rightarrow$ Sampling frequency.

Nyquist Rate, $f_N = f_s = 2f_{max}$

↳ minimum frequency



frequency যত বেশি accuracy তত বেশি।

Example 1.34: A signal having frequency ranges from DC to 10 KHz Determine the minimum sampling rate to recover the signal?

Soln: Frequency = DC to 10 KHz

= 0 to 10000 Hz

$f_{max} = 10000$ Hz

Minimum sampling frequency / Nyquist rate

$$f_N = 2 \times f_{max}$$

$$= 2 \times 10000$$

$$= 20000 \text{ Hz}$$

Example-1.35 : Consider the following signal

$$x(t) = 3 \cos 100\pi t$$

(a) Determine the minimum sampling rate to avoid aliasing

↳ alias মানে reconstruct perfectly না হলে

(b) Suppose the signal is sampled at $f_s = 200$ Hz, what is the discrete time signal after sampling.

(c) Repeat (b) if $f_s = 75$ Hz

(d) What is the frequency $0 < F < \frac{f_s}{2}$ of a ~~sine~~ sinusoid that yields samples identical to those obtained in part (c).

Soln: (a) $x(t) = 3\cos 100\pi t$ { $2\pi$ বরাবর থাকবে }

$$= 3 \cdot \cos . 2\pi . 50t$$

↳ এখানে $f_m$.

$$f_{max} = 50 \text{ Hz}$$

$f_N = 2 \times f_{max} = 2 \times 50 = 100$

$\therefore f_N = 100$ samples/sec or Hz

(a) যদি, $x(t) = 3\cos 100\pi t + 3\cos 900\pi t$

$= 3\cos 2\pi \cdot 50t + 3\cos \cdot 450 \cdot 2\pi t$

↳ $f_{max}$

এক্ষেত্রে সর্বোচ্চ যেটা max সেটা নিব।

$f_{max} = 450$ Hz

$\therefore f_N = 2 \times f_{max}$

$= 2 \times 450$ sample/sec

(b) $x(t) = 3\cos 2\pi \cdot 50t$ (continuous)

↳ discrete করলে $x(nT) = 3\cos 2\pi 50 \cdot nT$

T = sample time

T sec ↳ data 1

1 " " " $\frac{1}{T}$

$f_s = \frac{1}{T}$

$\therefore T = \frac{1}{f_s}$

$\therefore x(nT) = 3\cos 2\pi 50 \, n \frac{1}{F_s}$ | $T = \frac{1}{F_s}$

$= 3\cos \pi \cdot \frac{80}{200} \cdot n$ | $F_s = 200$

$= 3\cos \frac{\pi n}{2}$

$\therefore x(nT) = 3\cos \frac{\pi n}{2}$

(c) When $F_s = 75$ Hz

$\therefore x(nT) = 3\cos 2\pi 50 \cdot n \cdot \frac{1}{F_s}$

$= 3\cos 2 \cdot \pi \cdot \frac{50}{75} n$

$\therefore x(nT) = 3\cos \frac{4\pi n}{3}$

Example 1.36: The signal $x(t) = 20\cos(10\pi t)$ is sampled at a rate of 8 samples per second. Plot the amplitude spectrum for $|\Omega| \leq 30\pi$. Can the signal be recovered from samples?

Omega

(Ω/jΩ) ω বসাতে হবে

এখানে আছে, কারণ spectrum fourier transform ছাড়া হবে না

Sol$^n$: Here, $f_s = 8$

$$x(t) = 10\cos(10\pi t)$$

$$= 10\cos 2\pi.5t$$

$$f_{max} = 5$$

According to Nyquist theorem

$f_s$ must be an $f_s \geq 2f_{max}$

Now, $2f_{max} = 2 \times 5 = 10$

$\therefore f_s < 2f_{max}$

Hence the signal can never be recovered.

Example 1.37: A signal $x(t) = \text{sinc}(150\pi t)$ is sampled at a rate of (a) 100 Hz (b) 200 Hz (c) 300 Hz

For each of these three cases can the signal be recovered?

Soln: Given, $x(t) = \text{sinc}(150\pi t)$

$\Rightarrow x(t) = \text{sinc}\, 2\pi t \cdot 75$

$\therefore f_{max} = 75$ Hz

According to Nyquist Theorem,

$f_s$ must be an $f_s \geq 2f_{max}$

a) $f_s = 100$ Hz

$2f_{max} = 2 \times 75 = 150$ Hz

Here, $f_s < 2f_{max}$

Hence the signal can never be recovered.

(b) $f_s = 200\,Hz$, $2f_{max} = 2 \times 75 = 100\,Hz$

Here, $f_s > 2f_{max}$

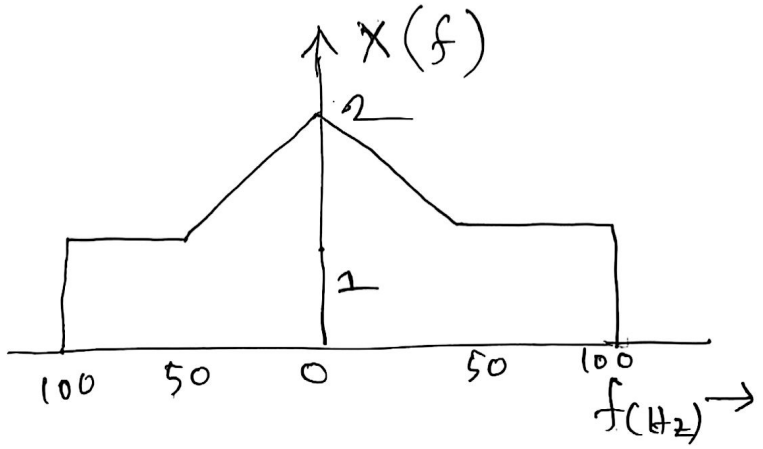
Hence the signal can be recovered.

(c) $f_s = 300\,Hz$, $2f_{max} = 2 \times 75 = 100\,Hz$

Here, $f_s > 2f_{max}$

Hence the signal can be recovered.

Example- 1.41 A signal $x(t)$ whose spectrum is shown $X(f) \rightarrow$ (fourrier transform) in below fig. sampled at a rate of 300 samples/s. What is the spectrum of the sampled discrete time signal?



Soln: Here, $f_m = 100Hz$

$$2f_m = (100 \times 2) = 200Hz$$

Sampling frequency, $f_s = 300Hz$

$$f_s > 2f_m$$

Hence there is no aliasing. the sketch is shown:

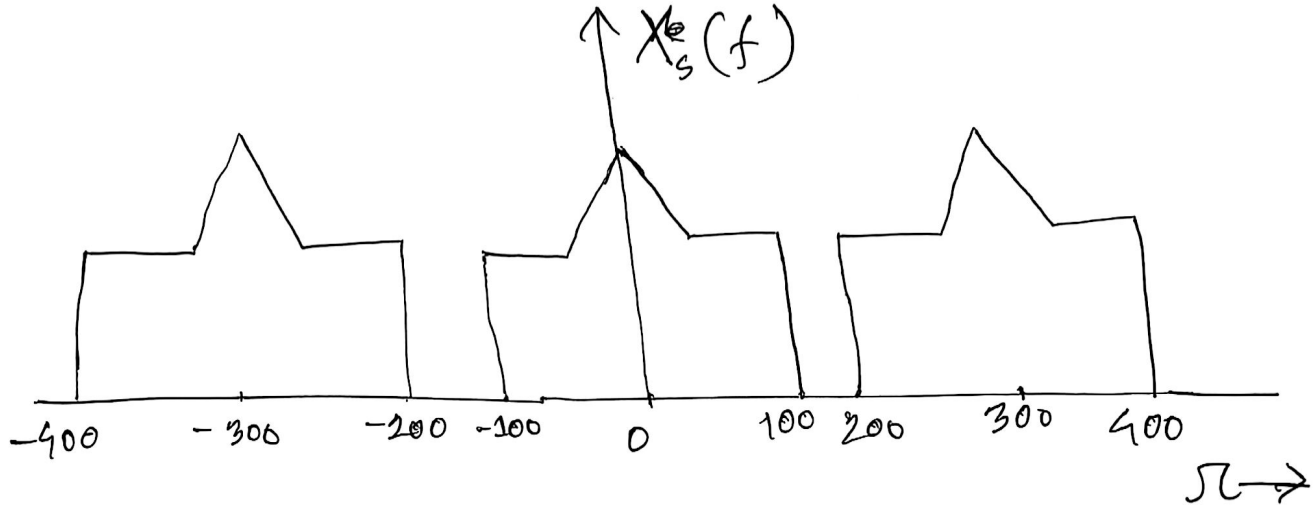



Fig: Spectrum of Sampled signal.

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## 1.8 Discrete Time System

Lecture-11
9.9.2026

### 1. Linear & Non-Linear:

Case:1 $y[n] = x[n]$

1. the sum of the weighted output:

$$y_3 = a_1 y_1[n] + a_2 y_2[n]$$

2. The ~~Sum~~ output due to sum of the weighted input

$$y_4 = a_1 x_1[n] + a_2 x_2[n]$$

if $y_3 \neq y_4$ then system is Non-Linear

and if $y_3 = y_4$ then system is Linear

Case-2:

$$y[n] = x^2[n] \longrightarrow \text{output} = (\text{Input})^2$$

$$\Rightarrow y[n] = \{x[n]\}^2$$

Let, $y_1[n] = \{x_1[n]\}^2$ & $y_2[n] = \{x_2[n]\}^2$

1. The sum of the weighted output:

$$y_3 = a_1 y_1[n] + a_2 y_2[n]$$

$$= a_1 x_1^2[n] + a_2 x_2^2[n]$$

2. The output due to sum of the weighted input:

$$y_4 = \{a_1 x_1[n] + a_2 x_2[n]\}^2$$

$$= a_1^2 x_1^2[n] + a_2^2 x_2^2[n] + 2a_1 a_2 x_1[n] x_2[n]$$

## 2. Time Invariant and Time Variant:

$$y[n] = n\,x[n]$$

Input shift: $y[n,k] = n\,x[n-k]$

Output shift: $y[n-k] = (n-k)\,x[n-k]$

here, $y[n,k] \neq y[n-k]$ ; time variant.

if, $y[n,k] = y[n-k]$ then system time ~~int~~ invariant

$y[n,k] \neq y[n-k]$ then system time variant

$y[n] = x[n] + x[n+1]$

Input shift: $y[n,k] = x[n-k] + x[n-k+1]$

Output shift: $y[n-k] = x[n-k] + x[n-k+1]$

here, $y[n,k] = y[n-k]$ ; time invariant

## 3. Causal and Non-Causal:

$$y[n] = x[n] + x[n-1]$$

when $n=0$, $y[0] = x[0] + x[-1]$

} Causal system

⇒ future context নাই

---

$$y[n] = x[n] + x[n+1]$$

when $n=0$, $y[0] = x[0] + x[1]$

} Non-causal

future context আছে

* Dynamic এর ক্ষেত্রে পরবর্তী future context এর দরকার ⇒ যাবে না।

## 4. Stable and Unstable:

$$y[n] = \sum_{n=0}^{\infty} u[n]$$

→ unstable system

⇒ infinite result তাই জন্য

$$y[n] = \sum_{n=0}^{4} u[n]$$

→ Stable system

⇒ finite result তাই জন্য

→ finite হলে stable.

→ infinite হলে unstable.

## 5. Static and Dynamic:

Static: → only present input

Dynamic: ⇒ Output → for past, future Input

present + past

$y[n] = x[n]$

$y[n] = x[n] + x[n-1]$

$y[n] = x[n+1]$

মনে রাখতে হবে,

0 ⟶ present

-1 ⟶ past

1 ⟶ future

## 6. Invertible and Inverse system:

$x[n] + y[n] \longrightarrow x[n]$